Are Immigrants Particularly Entrepreneurial? Firm Ownership as a Key Element in Immigrant Income Assimilation

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Abstract

Firm ownership is a defining feature of immigrant adaptation: 52% of immigrants own a firm at some point in their first 15 years post-arrival. We use Canadian data linking immigrant arrival records with individual and firm tax data to examine the role of firm ownership in adaptation to a host economy and the performance of immigrant owned firms. Higher immigrant firm ownership rates are mainly due to nonincorporated firm ownership, which looks like a last resort. Immigrant incorporated firms are smaller, invest less, and are less productive. There is little evidence in the data that immigrants are more entrepreneurial.

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1 Introduction

Firm ownership is a defining feature of the immigrant experience. In our Canadian administrative data, 32% of immigrant tax filers own a firm of some kind - incorporated or unincorporated, with or without employees - in their 15th year after arrival, and 52% have owned at least one firm at some time between arrival and their 15th year in Canada. In comparison, in the same data, 23% of our mainly native born comparison group own a firm of some kind in a given year. High rates of firm ownership plus evidence that the very process of migrating tends to select people who have lower risk aversion lead to the impression that immigrants are particularly entrepreneurial.(Jaeger et al. (2010),Gibson and McKenzie (2011))[1] Azoulay et al. (2022), for example, working with US data, conclude that, 'immigrants appear to 'create jobs' (expand labor demand) more than they 'take jobs' (expand labor supply) in the US economy.'(p.72) and call them 'highly entrepreneurial.'(p.84) Importantly, the perception that immigrants are entrepreneurial appears to drive policy decisions, as most major immigrant receiving countries have policies targeted at bringing in immigrant entrepreneurs (Fairlie and Lofstrom (2013)). For example, Canada's immigrant assessment system has included Investor and Entrepreneur classes since 1978 and the US has the E2 Visa for immigrants setting up and investing a significant amount of money in a new business.

In this paper, we investigate three inter-related questions: are immigrants are more entrepreneurial than the native born (in the sense of being more likely to open incorporated firms that employ others); do the entrepreneurs embedded in an immigrant arrival cohort account for at least as many jobs as there are workers in the cohort; and what role does firm ownership play in immigrant adaptation to the host economy? The answer to the first of these questions is of interest when considering the types of selection policies just mentioned. The answer to the second is of interest in relation to the large literature on the impact of immigrants on receiving labour markets. That impact is typically modelled in terms of the increased labour supply interacting with existing firms. There is, at times, discussion of whether immigrants either bring with them or attract new capital, but our approach is more specific in asking whether immigrants are creating jobs as well as providing new workers. The third question is of interest in its own right and is inter-twined with the other two because immigrant firms that are innovative and entrepreneurial both contribute to economic growth in the host economy and aid in successful adaptation for the immigrant owners.

There is a small literature on immigrant entrepreneurs that shows, among other results, that immigrants are more likely to own firms.(Fairlie and Lofstrom (2015)). We add to earlier

[1]Though that conclusion becomes more nuanced when considered at the family level (Dustmann et al. (2023))

work by virtue of our rich Canadian administrative data which consists of a linking together of the administrative records at time of arrival for all immigrants obtaining permanent residency between 2004 and 2019, the personal tax records for all those immigrants and for a native born comparison group, and firm tax records for all firms in Canada over the same years. One main advantage of this data is in measuring firm ownership. The firm tax records - which pertain to incorporated firms - include a schedule for privately held firms that lists all owners with at least a 10% share in the equity of the firm. From the personal tax records we get a record of whether individuals own a non-incorporated firm. Thus, firm ownership is clearly assigned for the whole population of immigrants who enter Canada in our data period. This is superior to US administrative data in which ownership has to be imputed(Kerr and Kerr (2016)). Fairlie and Lofstrom (2015) use the Survey of Business Opportunities in the US, obtaining special runs from the US Census Bureau on sales, employment, and payroll for immigrant and non-immigrant owned firms. Their data, like ours, show that immigrant owned firms tend to be smaller and pay less. Our data allows us to build on their results to examine dynamic patterns in terms of movements into firm ownership, firm growth and death, and to associate firm outcomes to characteristics of the owners. In addition, our data includes balance sheet measures that are the basis of our investigations into differences in investment and productivity both with time in the economy and in comparison to native born firms.

We use our data to examine immigrant firm ownership in three steps. The first is to document the importance of firm ownership in the immigrant experience. The combination of consistent definitions of ownership with complete longitudinal data allows us to show both cumulative and per-period ownership rates. It also allows us to break firm ownership into sub-categories. Levine and Rubinstein (2017) argue that it is important to delineate between incorporated and nonincorporated firms because the former are more likely to be based on an entrepreneurial idea while the latter may be a refuge of last resort for people who can't find regular employment. We focus attention on the earliest immigrant cohort we can observe in our data – immigrants whose first full year in Canada was 2005.[2] To provide a benchmark, we do the same for a native born 'cohort' of Canadian born people who were age 25 to 29 in 2005. The idea is to compare the immigrants to another set of people who were also entering on their economic life in Canada. We show that the immigrant cohort has higher firm ownership rates than the native born cohort almost from the outset. But we also show that the immigrant ownership advantage is mostly due to higher rates of self-employment with some extra ownership of incorporated firms with fewer than 2 workers. Ownership rates

[2]The patterns found for firm ownership in the cohort of 2005 are also present in other cohorts (see Appendix A).

of larger firms are quite similar between immigrants and the native born. We also present shifts in the distribution of market income (which includes both employment earnings and returns to capital) with time in the economy, showing that omitting firm owners and their incomes results in a distorted picture of immigrant income assimilation.

Given the importance of firm ownership for immigrants, in terms of both its prevalence and its role in income assimilation, we next turn to an analysis of the paths that the immigrant and native born cohorts follow through six states: paid employment; nonemployment; own account self-employment; ownership of an incorporated firm with fewer than 2 employees; incorporated firm ownership with 2 or more employees; and exiting the data. Following Humphries (2022), we implement a path clustering approach to delineate the most common paths through these states as the cohort entrants adapt to (or leave) the Canadian economy. The result is what we believe to be the most complete depiction of immigrant movements through employment, nonemployment, firm ownership, and exit from the labour market currently available.

A key point that emerges from the path analysis is that self-employment tends to be an absorbing state for immigrants, implying that it is not a stepping stone to owning a more "entrepreneurial" incorporated firm with employees. The immigrants who are in the predominantly self-employed path cluster also tend to be lower skilled, as measured by their education at arrival and their immigrant entry class. Following Harju et al. (2025), we examine movements in market income before and after entry into self-employment. Those patterns establish that the self-employed are negatively selected as measured by their income level and tend to be experiencing a downward trend in income and employment just before entering self-employment. Combining what we observe for the paths (the sequence of states that immigrants and non-immigrants), income dynamics preceding entry into self-employment, and the income levels of the self-employed, we argue that self-employment is, indeed, more accurately seen as a labour market state of last resort than part of an entrepreneurial journey.[3] Our comparisons show that this is particularly true for the immigrant cohort. Thus, our first main conclusion is that firm ownership plays a large role in immigrant adaptation to the host economy but that, for the most part, is not itself evidence that immigrants are more entrepreneurial. Instead, it reflects what might be construed as their need to use an

[3]These results fit with those in Georgarakos and Tatsiramos (2009) and Blume et al. (2009) which use panel data, from the US and Denmark, respectively, to examine immigrant entrepreneurship. Both find patterns suggestive of owning a firm being a stepping stone in a process of full integration into the host economy but both focus mainly on firm death and growth rather than firm creation. Also, neither paper distinguishes between incorporated and nonincorporated firm ownership, a distinction that we find is important. Hammarstedt and Miao (2018) investigate immigrant entrepreneurship using rich employer-employee matched data from Sweden but their focus is on who immigrant entrepreneurs employ.

accessible but poor labour market state to a greater degree than native born people entering the economy at the same time.[4] Indeed, while self-employment is a relatively low earning state, moving into it still appears rational since those who do so were on downward relative income path before opening their firm and that decline was arrested afterward. That might indicate that moving into self-employment generated better income than the owners would have faced if they had not opened their firm.

Immigrants, though, are still somewhat more likely to own incorporated firms and may create firms that are more successful in terms of employment growth, productivity, and investment. In our third step, we investigate each of those possibilities, again comparing outcomes for the 2005 cohort of immigrant entrants with the matching cohort of native born entrants. We begin by showing that, in contrast to the self-employed, incorporated firm owners are positively selected in terms of income levels and trends before firm opening, but incomes of immigrant firm owners are relatively weaker after firm opening than for their native born counterparts. This is a first piece of evidence that immigrant owned firms are relative under-performers – something that turns out to be the case in almost every dimension of firm performance. We show that immigrant firms are smaller on average, which is something that has been noted before (e.g., Fairlie and Lofstrom (2013), Kerr and Kerr (2020)). But we also show that the entire size distribution is lower for immigrant compared to native born owned firms, that immigrant firms have lower value added per worker (VAPW), lower investment, and are more likely to downsize or die. The only place they match the native born firms is in TFP growth, the distribution of which is very similar between native born and immigrant firms. The combination of these results may indicate that, in spite of the VAPW result, immigrant entrepreneurs are equally innovative, as measured by the growth they obtain with the same amount of capital and labour, but are constrained by lower amounts of investment. Notice, though, that even here, the argument is that they are at best equally productive. This is our second conclusion - that we cannot find evidence to indicate that immigrants are more entrepreneurial in the sense of creating firms that are more successful in terms of growth, productivity, or investment. Immigration - at least as Canada has implemented it - does not appear to be a route to raising the average entrepreneurial level of the host economy.

The relatively weak performance of immigrant firms translates into the number of jobs associated with them being far less than the number of workers entering in the same cohort

[4]The result that self-employment is effectively a weak labour market state is important for policy decisions given calls to consider counting Canadian experience in self-employment prior to permanent residency application as evidence of entrepreneurial ability ((Hiebert, 2020) citing conversations with the Economic Council of Canada).

with the entrepreneurs. One reason that immigrants are typically found to have little impact on the wages of native born workers might be that immigrant inflows have embedded in them enough entrepreneurs to keep ratios of entrepreneurs to workers in the economy constant. We do not find evidence to support such a claim and so the rationale for limited wage effects must lie elsewhere, in forces that are strong enough to overcome the implied reduction in the ratio of entrepreneurs to workers.

Our finding that immigrants are not substantially more entrepreneurial than the native born stands in contrast to the opposite claims made in several papers in the literature on immigrant entrepreneurs.[5] Most notably, Azoulay et al. (2022) use US administrative, survey, and scraped Fortune 500 data to argue that immigrants open more firms of all sizes, per capita, and that, by implication, the distribution of entrepreneurial ability is rightward shifted relative to that for the native born. Similarly, earlier papers point to greater firm ownership rates for immigrants as evidence of greater entrepreneurial contributions (e.g., Schuetze and Antecol (2005) and Lofstrom (2002)).

In part, our different conclusion stems from our differentiation between incorporated and nonincorporated firms since most earlier papers either don't make that distinction or actually focus on nonincorporated firm ownership for data reasons. Differences in conclusions may also come from our ability to directly observe ownership in a complete population of immigrants. We can't completely rule out the possibility that our different conclusions relative to some US papers comes from the US being a particularly attractive destination for more entrepreneurial immigrants (Kerr and Kerr (2020)) relative to Canada. But we view what we see as the most reliable results in Azoulay et al. (2022), the ones using the Survey of Business Opportunity, as also implying that immigrant firms tend to be smaller.

Our paper is most similar to Harju et al. (2025) who use similarly rich Finnish data to characterize selection into incorporated firm ownership and the performance of firms in Finland. We employ similar techniques but differ in our inclusion of self-employment firm ownership in our main discussion and in our focus on immigrant firm owners. As we will show, firm ownership plays a different role in immigrant adaptation to the economy than for the native born entering the economy at the same time. That different role - with more self-employment, a greater tendency not to exit self-employment, and poorer performance of incorporated firms - has implications for immigration policy. Our investigation also has similarities to Ranasinghe and Shao (2026) who use the same Canadian administration data. They mirror our results in terms of showing that immigrant owned firms under-perform

[5]Fairlie and Lofstrom (2013)provide a comprehensive review of these results and others related to immigrant entrepreneurship.

relative to native born owned firms and then calibrate a general equilibrium model. Using the model, they argue that the poorer performance of immigrant firms stems from distortions in accessing both capital and labour. We provide some reduced form evidence in the final section of the paper that reinforces their conclusions.

One possible caveat to our conclusion that immigrants are not better entrepreneurs relates to STEM workers and science and tech firms. There is a set of papers that focus on whether immigrants are more innovative in a sense captured by measures of patent filing, scientific citations and opening tech firms (e.g., Hunt and Gauthier-Loiselle (2010), Brown et al. (2019), Peri et al. (2015), Blit et al. (2020) Saxenian (2002)). This could be an alternative definition of being "entrepreneurial" (though one doesn't need to open a firm to make these types of contributions). That literature tends to focus on STEM workers and related industries and finds that immigrants make positive innovative contributions in those areas. Our results have nothing direct to say about innovative contributions of that kind, though we do show that firms opened by immigrants in the Science, Professional, and Technology industry are less productive and invest less than firms in the same industry opened by native born workers.

Our paper proceeds in five sections including the introduction. In the second section, we describe the data and present basic patterns. In section three, we examine the dynamics of movements across states, the characteristics of individuals following different employment/firm ownership paths, and income selectivity related to firm ownership, establishing that self-employment looks like a poor labour market state. In section four, we investigate incorporated firms performance measures. Section five contains conclusions.

2 Data and Core Patterns

At the heart of our investigation is a rich Canadian dataset matching workers and firms. This data is a version of the Canadian Employer-Employee Dynamics Database (CEEDD), which links together worker and firm tax records. In our special version, we further link in administrative data from the immigrant admission process. Our version of the CEEDD contains the universe of records for immigrant entrants back to 1982 and the linked worker-firm tax records for the years 2004 to 2019, inclusive, Here, we provide an overview of our data. A more complete description can be found in Green et al. (2016).

Firms are identified in the CEEDD using a unique identifier known as the Business Number (BN)[6] while workers are identified by their unique government identifier - the Social

[6]Not all businesses have a BN. However, they are required to have one if they have employees, sell goods

Insurance Number (SIN). BN's correspond to enterprise rather than establishment level units, thus a given BN may comprise several establishments. It is important to note that we are interested in whether individuals become firm owners rather than when a firm starts and so we track individuals not firms over time. On the worker side, the key forms are their personal income tax records (T1), the Financial Declaration, and the Business Declaration. From the T1, we get information on age, gender, earnings from paid work, non-work related income including dividends, and location of residence, which we code at the Census Division level.[7] The financial and business declaration files are constructed from tax forms filed by owners of nonincorporated businesses, and so it is from the personal tax records that we observe the creation of nonincorporated firms. As we will see, these nonincorporated firms rarely have employees other than the owner.

All personal tax records are compared to the Immigration Database (IMDB), which contains the administrative records for all permanent resident arrivals to Canada starting (in our version) in 1980, in order to identify immigrants. In order to administer the point system and other parts of the immigrant assessment system, the Canadian government gathers considerable data on new entrants. Among other data, the IMDB provides information on year of arrival,[8] age at arrival, source country, education at arrival, gender, marital status, and class of entry. We select individuals who filed a T1 at least once in our sample period. Apart from section 2.2, we do not distinguish between principle applicants (the person who was directly assessed for skills or investments in an immigrant admission case) and their accompanying family.

We are interested, in part, in understanding whether immigrants are more entrepreneurial than the native born. For that reason, we draw a comparison group consisting of native born Canadians who were age 25 to 29 in an immigrant cohort's first full year in Canada. We treat them as an 'entry cohort' entering the economy at the same time as the immigrants in the matching cohort. Following them provides a benchmark against which to judge whether increases in firm ownership after arrival for immigrants are similar to what the broader population goes through as it moves into economic activity or whether it is something special to immigrants. We chose age 25 as the lower bound for this group in order to make sure that

and services (and, therefore, have to charge the value added tax), or import or export goods and services. Thus, using the BN captures all firms in active economic activity.

[7]Census Divisions are geographic regions defined by Statistics Canada that are intermediate between provinces and municipalities. They are defined as groups of neighbouring municipalities grouped together for purposes of regional planning and managing common services. There are 293 Census Divisions in Canada.

[8]The year of arrival in our data is the year in which an individual becomes a permanent resident, which is not necessarily the year they first arrived in Canada. Since we are interested in immigrant firm ownership and non-permanent residents would have difficulty in borrowing money and with other legal arrangements to start a firm, we view the date of permanent residence as the relevant starting point.

they are (mainly) past their education years. In order to make the two groups comparable, we restrict our analysis to immigrants over age 24 at time of arrival since immigrants who were younger than that at arrival may have obtained some of their education in Canada, making them partly like other immigrants and partly like the native born.

For incorporated firms, the key administrative record is the T2 - the corporate income tax form. As part of this form, privately owned firms are required to file a Schedule 50 which lists all shareholders with at least 10% of common or preferred shares. The listing includes the shareholders' SIN's and it is through those that we can identify owners of incorporated firms, linking them to their individual tax records for all years and immigrant arrival records.[9] We follow immigrants from the time they enter the Canadian economy and say that they have become incorporated firm owners the first time they show up on a Schedule 50, even if the firm already existed or if there are multiple owners.[10] Finally, firm-level data is obtained from the National Accounts Longitudinal Microdata File (NALMF). This dataset includes relevant information about number of employees, total revenue, assets, value added, and several types of investment (e.g. machinery and equipment and on buildings) for the universe of incorporated businesses in Canada from 2000 to 2021 (Gu, 2024). We use the BN in the NALMF to link the firm information to owners via Schedule 50.[11]

The first step in our analysis is to determine the divisions of firm types we want to consider. Levine and Rubinstein (2017) examine the cognitive and non-cognitive traits that predict entrepreneurship, arguing strongly for the importance of separating incorporated from non-incorporated spells. They show that people who open incorporated firms tend to be rule breakers with high cognitive abilities while those who open non-incorporated firms have neither of these qualities. They hypothesize that risk takers incorporate exactly because they are taking a risk on a business idea and incorporation acts as a safe-guard for their own assets and for the firm. Non-incorporated firm owners tend to look more like they are making a choice of last resort rather than trying to pursue an innovative business idea. We will begin by employing their distinction but, later, we will provide evidence that key data patterns support the claim that these two types of firms are meaningfully different and,

[9]It is possible for firms to own shares in other firms. In those cases, we use the Schedule 50's of the owning firms to find the people who ultimately own them.

[10]For our data for immigrants entering in 2000, by 5 years after arrival, 78% of incorporated firm owners by this definition were sole owners and another 20% had only one partner.

[11]We want to focus on immigrants as entrepreneurs and so are not interested in ownership of shares in publicly traded firms. However, if firms start as privately owned but quickly go public then we could miss their existence in our data, which contains only privately owned corporations and nonincorporated firms. Fortunately, going public is very rare. Between 0.04% and 0.1% of privately owned firms go public in any year (Green et al., 2016). For privately owned incorporated firms started by immigrants after 2001, either 0, 1, or, rarely, 2 firms go public per year in our data period. If we observe a firm as private and then becoming public, we treat the public firm as a continuation of the private one.

therefore, that this is a useful way to categorize firms in our investigations. In addition to firm type, distinguishing between whether or not the firm provides employment is potentially important. The preference given to immigrant applicants who intend to open businesses in the immigration systems of most developed countries, including Canada, is supported by the claim that these immigrants will generate jobs - both for themselves and others. Some firms, though, are set up for tax related reasons and do not provide employment.

Based on this discussion, we see three potentially relevant firm ownership categories: incorporated firms with fewer than 2 employees (i.e., including firms with zero employees and ones that may only employ the owner); incorporated firms with 2 or more employees; and nonincorporated firms. We drop both incorporated and nonincorporated firms with zero net income on the assumption that such firms are dormant, referring to the remaining firms as 'active' firms.[12] For nonincorporated firms started by the 2005 immigrant entry cohort, only 2.7% had an employee other than the owner by 5 years after arrival (Green et al., 2016). Given this, we do not divide non-incorporated firms into sub-categories based on the number of employees. In contrast, the distinction between incorporated firms with fewer than 2 employees and those with 2 or more is more meaningful - on average, 63% of owners have fewer than 2 employees, while 38% have two or more. We therefore analyse both types of incorporated firms separately.

In assigning individuals to categories, we first categorize the person as an incorporated firm owner if she shows up as an owner on a Schedule 50 filed by a firm. Some of these owners may also be employed or own a non-incorporated firm but, given the substantial effort and cost of opening an incorporated firm, we see ownership of such a firm as a key defining event and list them only as firm owners. This will bias our incorporated firm owner numbers upward. We label the state of owning an incorporated firm with 2 or more employees as MAJINC (major incorporated firm) and the state of owning an Incorporated Firm with fewer than 2 employees as MICROINC. Next, we categorize people as Self Employed (SE) if their business income on their T1 tax form is greater than their employment income. The employed state (EMPL) includes all people who do not own an incorporated firm and who have earned income from a job that is greater than any nonincorporated business income. If an individual does not own an incorporated firm and does not report either earned or business income in a year (but does file taxes or is captured in some other tax form) then we classify them as nonemployed (NEMPL). And, finally, if tax data is missing in a given year then we categorize the person as out of the data (OUT). Including the OUT category allows us to look at attrition that may include re-migration and that has been shown to be

[12]This means we include firms with negative income since they may be active but performing poorly.

important in examinations of other outcomes for immigrants (Adda et al. (2022),Dustmann and Gorlach (2015)). It seems possible that difficulties in opening a business or a firm death could induce higher re-migration rates, and we want to capture those potential effects.

2.1 Time and Category Patterns

In Figure 1, we present patterns of total firm ownership with time since entry for both the immigrant and native born (NB) 2005 entry cohorts.[13] In Appendix A, we present the same plots for the 2010 and 2015 entry cohorts, showing that they follow the same substantive pattern during the years they are in our sample window. We define total firm ownership as the sum of the MAJINC, MICROINC and SE states plus ownership of nonincorporated firms in cases where the owner earns less from the firm than from employment. We express the ownership rates as proportions of the initial entrants in the cohort.[14] For comparison, the lone square marker at 2019 shows the proportion who own a firm among the native born of all ages (not just the 2005 cohort).[15] We represent movements in firm ownership in two ways. The pair of lines with circle markers present the proportions of each type of entrant who own a firm in a given year. In 2005, new entrant immigrants and the native born had roughly the same rate of firm ownership (11%). But that rate grew more rapidly for immigrants in virtually all the following years such that by 15 years after arrival, 29% of initial immigrant entrants were present in Canada and owned a firm versus 20% for the 2005 NB cohort. In comparison, the overall NB population had a 22% firm ownership rate. It is worth highlighting that we write these numbers as a proportion of everyone who 'enters' in 2005. It is more common to present these types of ownership rates as a proportion of the people present in the country in a year. Using that approach yields higher ownership rates (see Appendix A). We use initial entrants as the base to emphasize that re-migration is also an important part of the immigrant experience and because immigration policy is focused on what happens to the people initially admitted.

The lines with triangle markers in Figure 1 show the proportion of the two entrant groups who have ever owned a firm by each year after entry. These differ from the current

[13]We refer to cohorts by their first full year in Canada. Thus, the 2005 cohort consists of immigrants who obtained permanent residence some time in 2004. We do not use the part-year data from the year in which they actually entered.

[14]For this reason, the proportions at 15 years are lower in Figure 1 than the ones listed in the introduction, which are proportions of people present and filing taxes in year 15.

[15]Because our immigrant data begins in 1980, we cannot separate immigrants who arrived before that date from the native born. Using data from the 2011 National Household Survey, immigrants who arrived before 1980 made up 7% of the total of those immigrants plus the native born in that year. Thus, the all ages NB group consists primarily of the native born. The 2005 native born cohort is not affected by this problem and consists only of native born in that age cohort.

ownership lines to the extent that some people move in and out of firm ownership. The upper, immigrant cohort, line is striking. By 15 years after arrival, 46% of immigrants have owned a firm at some point. Firm ownership is clearly a central feature of the immigrant experience. Interestingly, it is also important (though somewhat less so) in the lives of the recent generation of the native born. By 15 years after what we define as entry for this group, 35% have owned a firm of some kind at some point.

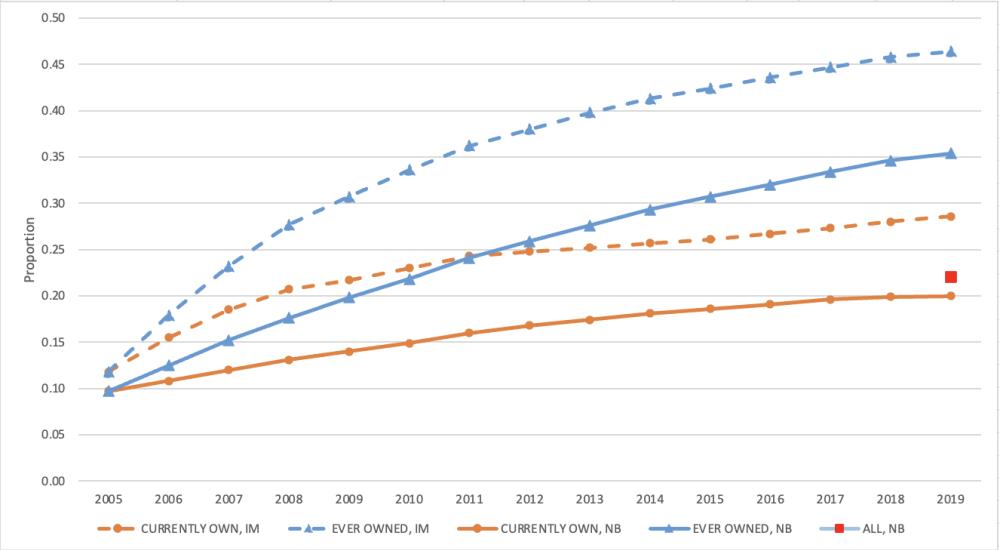


Figure 1: Proportion of People Who Currently Own or Have Ever Owned a Firm By Year: 2004 Immigrant and Native Born Cohorts

In Figure 2, we plot current firm ownership against years since arrival broken down into our different firm-type categories.[16] Immigrants and NB have similar proportions who own incorporated firms of both types in their first full year after entry, with immigrants having higher SE rates. In the following years, immigrant ownership exceeds that for the comparison NB group in all the categories. Importantly, the finding from Figure 1 that immigrants are substantially more likely to own firms soon after entry arises mainly from nonincorporated firm ownership, with some contribution of MICROINC firm ownership. By 5 years after arrival, 8.4% of immigrants own an SE firm compared to 4.5% for NB, with the differential shrinking only slightly over the next 10 years. In contrast, by 15 years after arrival immigrants have only slightly higher ownership rates of MAJINC firms (3.6% versus 3.2%). For both immigrants and NB, the rates of ownership of this type of firm - likely the type that governments are hoping immigrant entrepreneurs will open - is small and increasing

[16]Note that these proportions do not add up to the proportions in Figure 1 because people who own non-incorporated firms but get more than half their income from paid earnings are not included in any of our categories.

only slowly. Thus, the immigrant advantage in firm ownership is mainly due to their having more firms without other employees, not more ownership of firms that are more properly described as entrepreneurial.[17]

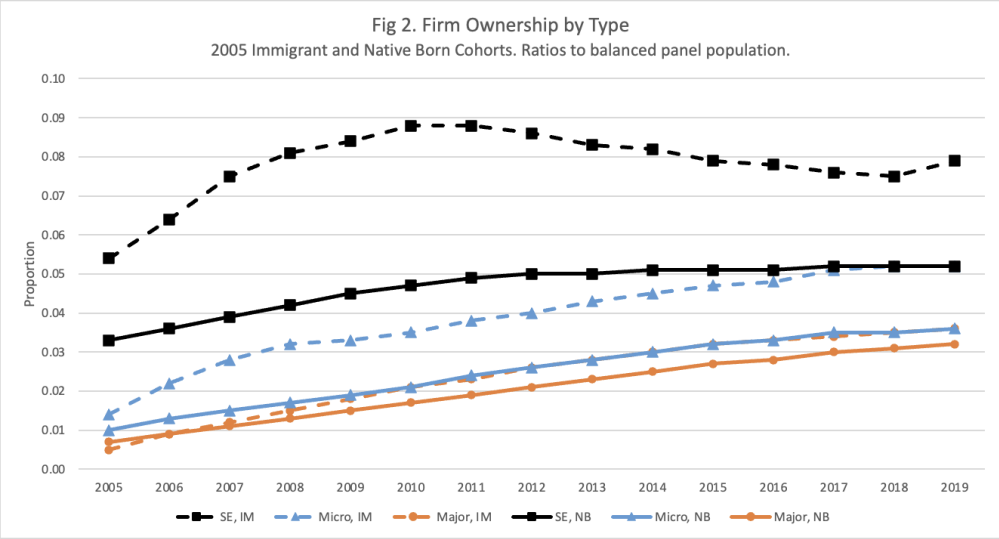


Figure 2: Firm Ownership by Type: 2004 Immigrant and Native Born Cohorts

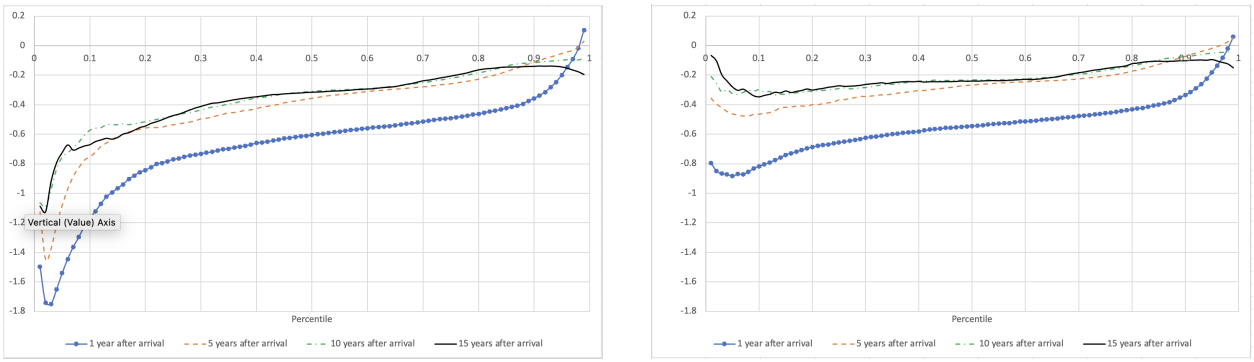
The findings that immigrant firm ownership rates rise with time in Canada and ultimately surpass those of the NB fit with results in earlier studies (Li (2001), Schuetze and Antecol (2005), Frenette (2004) and Hou and Wang (2011)). All of these papers focus on the dual questions of the proportion of immigrants who become self-employed (i.e., own nonincorporated firms) and their earnings. All except Li (2001) use Census data where firm earnings are only collected for nonincorporated firms and so their findings pertain only to those firms. Li (2001) uses tax data but also only examines non-incorporated firm ownership.

2.2 Income Assimilation

The results that immigrants are more likely to own a firm and that firm ownership forms such a large part of the immigrant experience imply that ownership income could play an important role in the immigrant income assimilation experience. The common approach of

[17]In Appendix A, we investigate whether higher immigrant firm ownership and any differences in firm performance might be related to the fact that immigrants enter through a selection system that includes specific classes for entrepreneurs while the native born do not. We show that of the immigrants who are ever in the MAJINC state in their first 15 years after arrival, less than 5% entered through the entrepreneur entry class. The rest were either selected for broad labour market skills or entered as refugees or accompanying family members and were not selected for economic relevant attributes at all. Given this, the vast majority of immigrant firm owners are no more selected for entrepreneurship than NB entrants and, so, we would not expect their firms to perform better because of greater selectivity.

dropping firm owners and focusing only on paid employment earnings could then paint quite a different picture relative to the immigrant experience in terms of their total income.[18] To investigate the extent to which this is true, in Figure 3 we plot the differences in log percentiles (immigrant minus NB) for our immigrant and NB 2005 entry cohorts for all percentiles between the 2nd and the 98th for 1, 5, 10 and 15 years after Canadian labour market entry. The left panel presents these differences for market income, i.e., the sum of income from all sources other than transfer income. This includes paid employment earnings, self-employment earnings, and income from investments and rents. The right panel presents the same differences but restricted to paid employment earnings. In both cases, we restrict the analysis to the balanced panels of people who are present in all 15 years in order to focus on actual assimilation paths and avoid patterns induced by selective emigration.(Lubotsky (2007), Rho and Sanders (2021), Akee and Jones (2026)) We also restrict attention to those with positive income of the relevant type.



(a) Market Income

(b) Paid Employment Income

Figure 3: Immigrant - Native Born Log Income Differences, by Percentile, across Years Since Entry

In the first year after arrival, immigrant incomes of both types are considerably lower than the incomes of the NB entry cohort. Median market income is .6 log points lower for immigrants than the NB while median employment income is .55 log points lower. But the figures make it evident that the income differential facing immigrants is much larger in the lower tail of market income than employment income. At the 10th percentile, the immigrant disadvantage in market income amounts to -1.17 log points while their disadvantage in employment income is -.8 log points. Over time, immigrants close the gap to some extent but, again, the lower and upper parts of the distribution tell different stories. The immigrant market income distribution, relative to the NB distribution at each point after entry, shifts

[18] Among recent papers, Borjas (1986), Rho and Sanders (2021), Dustmann et al. (2024), and Albert et al. (2026) focus only on the paid employed.

up more or less in parallel, maintaining the extra disadvantage at lower percentiles. For employment income, instead, the differential line rotates up at the bottom tail, resulting in a relatively uniform disadvantage of around .25 log points up to about the 70th percentile. In a counterfactual exercise (Appendix A), we show that the different patterns in the lower tail of the market income distribution arise because of the much higher proportion of immigrants who are SE. The fact that SE is concentrated in the lower half of the overall market income distribution accounts for the finding that the differentials in the upper parts of the market and paid employment distribution move similarly while the lower parts follow different patterns. But the key implication is that firm ownership - which we have seen is such an important part of the immigrant experience - is actually a drag on immigrant incomes such that including it in our analyses results in a depiction with much worse immigrant assimilation, especially in the lower tail of the income distribution.

We are not the first to include earnings from firm ownership in as assessment of immigrant assimilation. Baker and Benjamin (1994), using Canadian Census data, include self-employment earnings. More recently, Lubotsky (2007), Akee and Jones (2026), and Villarreal and Tamborini (2018) include self-employed earnings in studies using longitudinal US administrative data. Our data bring several advantages over these studies. Because the US administrative data does not include immigrant status, these studies need to work with linkages to survey data, with linkage rates as low as 80%. We have the universe of immigrants identified by their arrival records. These studies also only have information on income reported as labour earnings while we are able to include those earnings plus capital earnings from the firm. This provides a more complete picture of the role of firm ownership in immigrant income progression. Also, while some previous studies include self-employment earnings in their total earnings measure, they do not investigate the separate roles of firm ownership income and paid earnings in the immigrant experience. The only exception to this is Villarreal and Tamborini (2026) which uses linked CPS and tax data for the US to investigate the rate of self-employment and patterns of self-employment earnings with time in the economy. Our results showing larger self-employment rates for immigrants mirror their findings, and they also show that assimilation is slower when measured including self-employment earnings. Their data does not include incorporated firm income and they focus on mean log earnings, missing the substantially different pattern across the income distribution that we document.

These figures highlight the need to understand both the pathways into these different firm ownership states and how the firms that are opened perform.

3 Dynamics of Movements Across States

The fact that the current year and ever-owned proportions are not identical in Figure 1 implies that there are shifts in and out of firm ownership over time. There are also potential shifts across the various firm categories. Studying those movements can both provide more insight into the nature of the categories and some further understanding of the extent of entrepreneurship. For example, if we find that owners of SE firms have a high probability of transitioning to the ownership of MAJINC firms then the high proportion of SE's among immigrants could be seen as a hopeful sign. This is a descriptive exercise not a causal one. That is, we are interested in documenting the patterns that immigrants typically follow rather than, say, whether a policy that increased employment would cause an increase in firm opening.

3.1 Clustering Paths

We characterize movements across our six states (OUT, NEMPL, EMPL, SE, MICROINC, and MAJINC) for each individual in our 2005 immigrant and native born cohorts over a 15 year period (2005-2019). Our goal is to document the most common paths, the individual characteristics associated with each, and how both movements and characteristics differ between immigrant and native born market entrants. Our approach broadly follows that in Humphries (2022) who characterizes life-cycle firm ownership and employment paths using rich Swedish data. He searches for a set of the most common paths using an algorithm with two steps. The first consists of generating a 'distance' between every pair of paths (i.e., pairs of people) based on a notion of the cost of converting one of the paths to another, with the cost being zero if the paths are identical and higher the more deletions and switches that have to be made in different years to convert one to the other. The second involves using a clustering algorithm to generate clusters of paths with minimal distances among the various specific paths within each cluster. As he demonstrates, this approach can identify clusters with different dynamic patterns in, say, moving between employment and incorporated firm ownership. The difficulty with this approach is that paths across discrete states like ours can vary in terms of durations in each state, ordering of sequences, and timing, and different algorithms for deciding which sequences should be treated as most similar are quite complex and can generate quite different clusters (Studer and Ritschard (2016)).

In response to these issues, we adopt a simpler, transparent approach. First, we create

two vectors for each person:

$$X_i = (N_{i,OUT,2005-2011}, N_{i,NEMPL,2005-2011}, ..., N_{i,MAJINC,2005-2011}) \in \{0, 1, ..., 7\}^6,$$

and

$$Z_i = (N_{i,OUT,2012-2019}, N_{i,NEMPL,2012-2019}, ..., N_{i,MAJINC,2012-2019}) \in \{0, 1, ..., 8\}^6,$$

where N_{i,m,t_0-t_1} denotes the number of years individual i spends in state m over the period from t_0 to t_1. Then, for each individual, we stack these two vectors into $Y_i = (X_i, Z_i)$ and search for common patterns in the data using a clustering algorithm. In particular, we use the K-means algorithm in which we pre-specify a fixed number of clusters, K, and assign observations to the clusters in such a way that, within each cluster, the average dissimilarity between the observations and the cluster-specific mean is minimized (Hastie et al., 2009). To determine the optimal number of clusters, we estimate K-means with different values of K (ranging from 2 to 20), following the approach of Audoly et al. (2024) and Makles (2012)in computing an R^2 measure on each iteration.[19] We select the optimal number of clusters, K^*, as the one that maximizes the R^2 measure while also yielding a meaningful improvement (of at least 5%) relative to the variance explained by the data using $K^* - 1$ clusters.

We chose this approach because duration is a key dimension for several of our states. For incorporated firm ownership, creating firms that last is important for contributions to economic growth and a measure of the success of an entrepreneur. On the other hand, to the extent that owning an SE firm is a potentially problematic state that represents an inability to either find paid employment or open an incorporated firm, longer duration in this state would be a marker of difficulties in adaptation to the economy. The same holds for the NEMPL state. At the same time, we are concerned with sequencing, and grouping by durations alone would take no account of how a person entered the state or whether the total duration consists of multiple, shorter spells. We allow clusters to be defined by sequence, to some degree, by including durations in both the first seven and second eight years after

[19] In particular, the measure is:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(Y_i - \bar{Y}_{C_{j(i)}})^2}{\sum_{i=1}^{n}(Y_i - \bar{Y})^2},$$

where $C_{j(i)}$ denotes the cluster to which individual i is assigned, $\bar{Y}_{C_{j(i)}}$ is the average outcome in cluster $C_{j(i)}$, and $\bar{Y}$ is the overall mean. Intuitively, this metric captures the share of the variation in individual trajectories explained by the cluster average trajectory (Audoly et al., 2024). Makles (2012) denotes the R^2 measure as η^2, and describes it as capturing the proportional reduction of the within sum of squares compared to the total sum of squares.

entry. This, for example, would allow for there to be a common group of paths consisting of extended SE spells as a precursor to opening a successful incorporated firm. We will also investigate sequencing further by presenting graphs of the year-by-year paths followed by the people grouped in each cluster.

Following Humphries (2022), we begin in Figure 4 with a graph of the paths for all the people in our 2005 immigrant entry cohort, with colour coding of the six states.[20] The most dominant path, with about a quarter of entrants, is to enter and stay in employment throughout the 15 years. A further 50% move into paid employment as an absorbing state such that by 2019, 71% of the original entrants are in paid employment. But, interestingly, only 17% follow simple paths with non-employment followed directly by a permanent move into employment. These are the types of paths typically emphasized in the existing literature on immigrant adaptation to the host economy since both people owning firms and those who are not in the data for periods after arrival are not included in standard depictions. Approximately 5% of the 2005 entrants are only observed in the OUT or NEMPL state. Taking them as people who either leave Canada or never engage in economic activity in Canada, people who are only ever in the EMPL or NEMPL states make up 50% of the remaining people - the people who are ever active in Canada. The remainder either have some spells out of the data or of firm ownership. Paths that include some firm ownership before entering employment make up 10% of all those who flow into permanent employment.

Entering SE as an absorbing state becomes an increasingly common path as time progresses, with 3% of immigrant entrants entering SE and not leaving it during our 15 year period. It is also somewhat common to see an SE spell just before a person enters employment. But there is little evidence of an SE spell preceding a MAJINC spell and only limited evidence of SE to MICROINC moves. In contrast, there is evidence of movements from MICROINC to MAJINC, particularly toward the end of our period, with movements from EMPL to MAJINC being more common in the first decade after arrival.

[20]For confidentiality reasons, we drop any paths followed by fewer than 10 people.

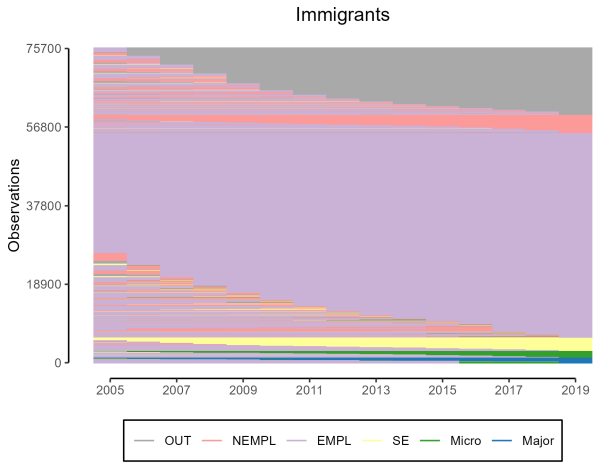


Figure 4: Paths Across States with Time Since Entry: Immigrants

Figure 5 shows the complete set of paths for our 2005 native born cohort. The figure looks broadly similar to Figure 4 but on closer inspection, a larger proportion of the native born enter the EMPL state and stay in it throughout the 15 years (43% for the native born versus 28% for immigrants). In part, this is likely because calling the ages of 25 to 29 'entry' is a bit of a fiction - some of the people in the 2005 native born cohort will have actually entered the active economy some years earlier. The main difference between the two groups is the larger proportion of immigrants who leave the data (and, potentially, the country). This is as one might expect, though it is still the case that 16% of the native born have left the data by 2019. Permanent movements into the SE state are larger for immigrants while the patterns for MICROINC and MAJINC firm ownership look quite similar - all of which fits well with Figure 2.

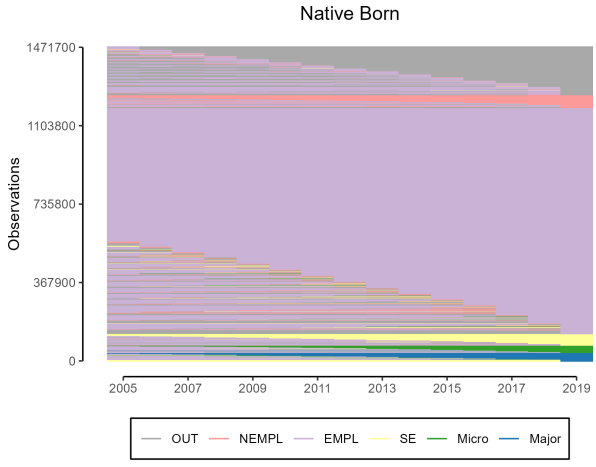


Figure 5: Paths Across States with Time Since Entry: Native Born

Our clustering approach results in these figures being broken down into 6 clusters.[21] We present the path plots for each of these in Figure 6 for the immigrant cohort. Examining the various plots allows us to see what the paths within each one have in common. The six clusters (with the percentage of the 2005 immigrant entrants in each one) are as follows.

Leavers (18.4% of entrants) The first cluster consists mainly of paths that end in persistently remaining in the OUT state. The figure shows that many of these OUT spells were preceded by EMPL spells and, in fact, on average across our time period, 80% of moves into the OUT state came from an EMPL spell. Since 70% of all entrants were in the EMPL state at the mid-point of our data (year 7), this implies that employment is a disproportionately common path to leaving the data. NEMPL is also disproportionate, accounting for 13% of movements into the OUT state in this cluster but 9% of the sample at 7 years, as is SE, accounting for 5% of movements into OUT and 4% of the sample. The incorporated states are both under-represented, which potentially fits with the point in Adda et al. (2022) that choosing not to re-migrate is associated with choices to make greater investments in the host country, though in their case the investment related to human capital rather than physical capital in a firm.

Late Entrants (11.6% of entrants) This cluster primarily contains paths that start at NEMPL, OUT, SE, or EMPL, then experience a period of NEMPL, and end up moving into EMPL permanently. Here, there is no evidence of frequent paths related to MAJINC ownership, suggesting that starting in NEMPL is not a strong predictor of entrepreneurial activity in subsequent years.

Incorporated (7.9% of entrants) The third cluster contains paths ending in either MICROINC or MAJINC spells. It is interesting that these spells are grouped together - there is little movement between MICROINC AND MAJINC spells in the second 7 year period so it seems possible the algorithm would want to give each their own cluster. But they share roughly similar times in EMPL and limited time in SE in the first 7 years. A main difference between the MICROINC and MAJINC ending spells is that the former includes a noticeable number of paths that include the SE state while the MAJINC paths have almost none. The latter is true even in the indirect sense - many MAJINC spells are preceded by MICROINC spells but those specific MICROINC spells are not preceded by SE spells. That is, there is a path from SE to MICROINC, with the latter as an absorbing state, but not a common SE to MICROINC to MAJINC path. This demonstrates the strength of the clustering approach - allowing us to summarize key patterns. In this case, it indicates that SE is not a step in a

[21] With six clusters, we capture 69.3% of the total variation in the data, representing an increase of 5.2% over the R^2 measure with five clusters. In contrast, increasing the number of clusters to seven yields only a marginal gain in explained variance of 3.8%.

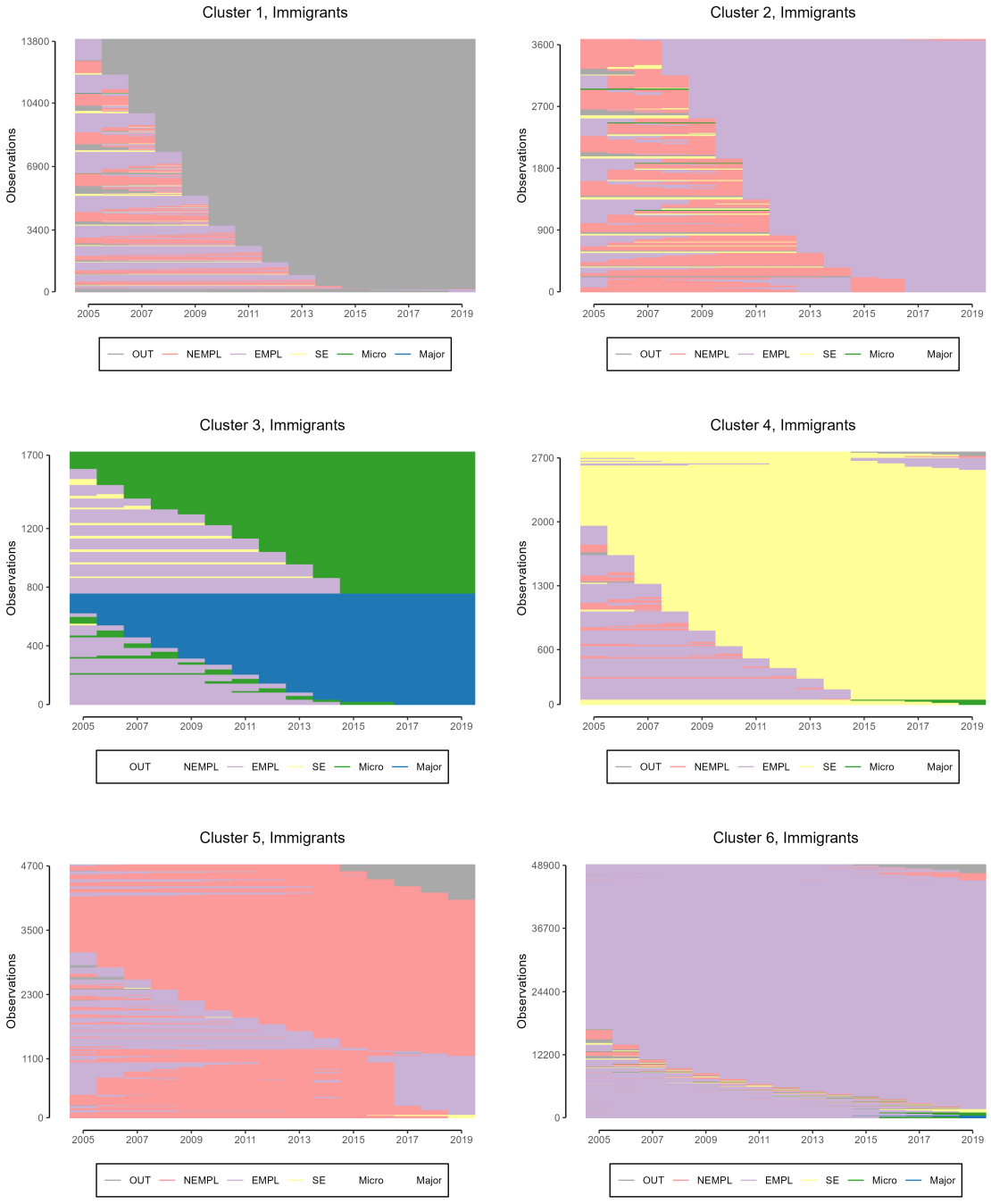


Figure 6: Paths Across States with Time Since Entry by Cluster: Immigrants

move toward opening truly entrepreneurial (MAJINC) firms.

Self-employed (7.6% of entrants) The fourth cluster consists of people who spend protracted periods in the SE state, many of them experiencing it as an absorbing state. The most common state just before entering SE is EMPL, but NEMPL also makes up a disproportionate share. Interestingly, this group shows little tendency to move in and out of SE - it is a group who adopt it as their main economic activity and stick with it. In 2019, 2.4% of this group were in the MAJINC state, which is less than half the proportion of the entry cohort who were in that state in that year. The fact that the algorithm groups SE dominant paths separately from paths dominated by incorporated firm ownership supports the claim in Levine and Rubinstein (2017) that incorporated and non-incorporated firm ownership are meaningfully different states. This separation likely arises because SE tends to be an absorbing state rather than an entryway to incorporated firm ownership.

Non-employed (11.1% of entrants) This cluster consists of paths with considerable NEMPL spells and can be divided into a group who (eventually) move into NEMPL as an absorbing state and another group who move back and forth between EMPL and NEMPL but end up in EMPL, with a small group ending up leaving the data and another small group ending up in SE. It is noteworthy that there are few SE spells here. To the extent that SE spells start or end with EMPL or NEMPL spells, that seems to mainly happen with those labour force spells ending up in SE (i.e., to be found in the SE cluster) rather than the other way around.

Paid Employed (43.5% of entrants) The sixth and largest cluster consists of paths with extended spells of EMPL, with almost all the paths being in the EMPL state by the end of our period. The majority of the paths are in EMPL throughout, but there is a substantial proportion that has an initial period moving back and forth between EMPL and other states. SE is in evidence to a limited extent in these early years. Together with the Non-employed and Self-employed group patterns, these results indicate that SE is less a first step in a move toward paid employment and more an absorbing state.

We present the plots of paths by cluster for the native born in Appendix A. We selected the number of clusters in the immigrant cohort exercise and used that number for the NB cohort but otherwise, it is the NB data determining what to group together into the six groups. It turns out that it chooses six remarkably similar groups and we will use the same names to describe them. The percentages in each group are: Leavers (11.5%); Late Entrants (9.4%); Incorporated (5.9%); Self-employed (5.0%); Non-employed (7.1%); and Paid Employed (61.0%). Thus, the NB cohort has much larger percentages in the Paid Employed group and much less in the Leavers and Non-employed groups. The path patterns in the NEMPL and EMPL groups are substantially the same for immigrants and NB. In

contrast, the Leavers group shows more movements in and out of the data for the NB while OUT is more often an absorbing state for immigrants, perhaps reflecting that it is capturing re-migration for immigrants. In addition, for the NB, the INC group shows more paths in which SE leads directly to MAJINC and an SE to MICROINC to MAJINC path is more in evidence. Furthermore, the SE group for the NB shows more movements in and out of SE status. Thus, our conclusions about the SE state are somewhat different for the native born: rather than something that is often an absorbing state and rarely leads to opening a MAJINC firm, for the native born, it is a more fluid state and, for some, is a stepping stone in an entrepreneurial path.

We can summarize these patterns, to some extent, using the transition matrix among the states that underlie our clusters. In Table 1, we present the transition matrix for moves among states between year 5 and year 6 after entry, separately for the immigrant and NB 2005 entry cohorts. The states are the ones defined earlier: OUT, NEMPL, EMPL, SE, MICROINC and MAJINC. Each cell in the table is read with each row corresponding to a person's state in year t and the row entries showing the probability a person moves to the state in the column dimension by year t+1. For example, the probability an immigrant who is not employed (NEMPL) in 2009 moves to EMPL in 2010 is 0.205. We present the transition matrices for other years in Appendix A. The matrices for later years are quite similar to the one presented here. The matrices for the first few years have higher probabilities of immigrants moving from EMPL to NEMPL.

The immigrant panel reveals that the probabilities associated with staying in and moving between the employed and nonemployed states are high. Approximately 87% of the nonemployed in year 5 are still either employed or nonemployed in year 6, with about a fifth of them moving into employment. On the other hand, the rates of transition into ownership of an incorporated firm of either type from employment or nonemployment are relatively low, being below 1% in all cases. The flows from SE to incorporated firm ownership are decidedly larger than from the NEMPL and EMPL states but are still small, with less than 1% opening a MAJINC firm. Less than 10% of new MAJINC firms were opened by a person who owned an SE firm in the previous year. The probability of moving from SE to MICROINC is higher, as we saw in our cluster analysis, but still not large. And the probability an SE person moves into either employment or non-employment is over five times higher than the probability that they open an incorporated firm. If we group states together based on patterns of movements among them, a grouping with EMPL, NEMPL, and SE is much more reasonable than grouping SE with MICROINC and MAJINC. That is, own-account, non-incorporated firm ownership looks more like another employment state than part of an entrepreneurial process. Echoing results in Gendron-Carrier (2018), who works with Canadian tax data not

Table 1: Transition Matrices: Year 5 to Year 6 After Entry

2009			**Immigrants**			
State at	year t+1					
year t	OUT	NEMPL	EMPL	SE	MICROINC	MAJINC
OUT	**0.830**	0.040	0.091	0.029	0.008	0.003
NEMPL	0.069	**0.662**	0.205	0.055	0.007	0.002
EMPL	0.035	0.060	**0.870**	0.024	0.009	0.002
SE	0.060	0.064	0.128	**0.715**	0.027	0.007
MICROINC	0.038	0.025	0.077	0.029	**0.711**	0.120
MAJINC	0.028	0.013	0.038	0.009	0.113	**0.798**

2009			**Native Born**			
State at	year t+1					
year t	OUT	NEMPL	EMPL	SE	MICROINC	MAJINC
OUT	**0.765**	0.031	0.179	0.017	0.005	0.003
NEMPL	0.074	**0.661**	0.226	0.035	0.004	0.001
EMPL	0.054	0.030	**0.897**	0.013	0.005	0.002
SE	0.074	0.042	0.132	**0.722**	0.024	0.006
MICROINC	0.052	0.009	0.073	0.012	**0.742**	0.112
MAJINC	0.032	0.003	0.030	0.004	0.079	**0.853**

broken down by immigrant status, only 19% of the new firm starts by individuals who were nonemployed in the previous year are incorporated firms compared to 30% among those who were previously employed. This, too, suggests that SE is a lower quality state and fits with the division of firm ownership spells into 'necessity' and 'opportunity' spells in other papers (Block and Sandner (2009),Humphries (2022)). In contrast, 12% of MICROINC firms convert to MAJINC, indicating that opening an incorporated firm with no employees can be a step toward a more substantial firm. This fits with our cluster paths analysis, which adds the detail that even though SE to MICROINC moves are not uncommon, this is not a first step toward MAJINC ownership. In addition, immigrants tend to move into SE from other employment states but not to move out again. Finally, the lowest rates of leaving the data (and, possibly, the country) are for owners of incorporated firms.

The second panel shows the same transition matrix for NB new entrants, and it is striking for its similarities to the immigrant matrix. Movements from SE to owning an incorporated firm are even smaller, with similar sized flows into EMPL and even larger flows out of the data. The flow rates from NEMPL to EMPL are similar for the two groups, though immigrants show a much higher rate of moving from EMPL to NEMPL. The flow rates from all states out of the data are surprisingly larger for the NB than immigrants, though the ordering of the states from which this is more likely to happen stays largely the same.

3.2 Characteristics of People in the Clusters

In Table 2, we present characteristics at time of arrival for immigrants in the various clusters. The Incorporated and Paid Employed clusters have quite similar – and relatively high – education levels with only 17% of the former and 20% of the latter having a high school education or less. The Incorporated cluster is more likely to have entered in the Skilled Worker class, and, so, assessed based on skills like education, but only a bit more likely to have entered in the Entrepreneur or Investor classes than other clusters apart from the Paid Employed. The Self-Employed, on the other hand, have lower education than the Paid employed and are more likely to have entered in the Family or Refugee classes, which are not selected for skills. The Non-employed stand out for being more female and lower education. In a number of ways, the Self-Employed are more similar to the Non-employed than to either the Paid Employed or the Incorporated clusters.

Table 2: Characteristics by Cluster at Time of Entry: Immigrant Cohort

			Cluster			
	Leavers	**Late Entrants**	**Incorp**	**Self Empl**	**Non-Empl**	**Paid Empl**
Cluster Proportion	0.184	0.116	0.079	0.076	0.111	0.435
			Proportion Within Cluster			
Female	0.476	0.666	0.356	0.425	0.704	0.503
Married	0.742	0.821	0.809	0.806	0.833	0.749
Education						
Secondary or less	0.208	0.25	0.173	0.26	0.342	0.197
Some postsec	0.157	0.185	0.165	0.212	0.171	0.184
Bachelor degree	0.404	0.387	0.447	0.375	0.318	0.439
Master's	0.172	0.136	0.175	0.121	0.104	0.143
Doctorate	0.033	0.016	0.031	0.017	0.018	0.021
Entry Class						
Skilled workers	0.481	0.473	0.571	0.446	0.33	0.53
Entrepreneurs	0.012	0.012	0.019	0.013	0.013	0.003
Investors	0.039	0.017	0.022	0.018	0.076	0.004
Self-employed	0.005	0.004	0.007	0.013	0.005	0.002
Provincial nominees	0.02	0.014	0.038	0.023	0.018	0.029
Sponsored family	0.335	0.296	0.229	0.278	0.329	0.246
Refugees	0.103	0.176	0.109	0.198	0.223	0.145

In Figure 7, we plot mean market income (income from all sources other than transfers, including earned income and dividends) for people in each cluster. These are annual incomes, including zeros, and will reflect nonemployment, part year or part time work patterns. They will also reflect changing selection across states over time. For example, in the Self-Employed cluster in the early years, the cluster members were mainly in paid employment but later moved primarily to self-employment. The Incorporated cluster is the top income group for both immigrants and NB, which is suggestive of positive selection into opening incorporated firms – a point we return to in section 3.4. But the time paths of market income in the Incorporated cluster are noticeably different between immigrants and NB, following a path that soars above the (itself increasing) income path for the Employed cluster for the NB but runs parallel to that path for immigrants.[22] This is a first indication of a theme that will run through the remainder of our investigations: immigrant owned incorporated firms are under-performers relative to those owned by the NB.

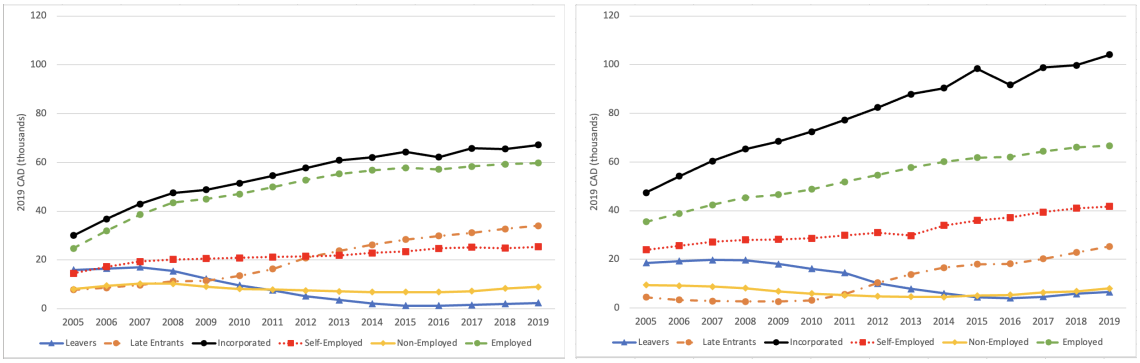
The Self-Employed cluster income paths are quite flat for both immigrant and NB owners but, again, flatter for immigrants. They also are much lower than either the Employed or Incorporated clusters, fitting with negative selection that we investigate in section 3.3 and with our ongoing narrative that self-employment is more a failed state than an entrepreneurial one. This echoes a similar finding in Humphries (2022) that the self-employed have lower earning potential than incorporated firm owners. Using Swedish longitudinal data, he also finds that the own account self employed are meaningfully different from incorporated firm owners, with the latter having more education, higher earnings, and higher cognitive and non-cognitive measures.[23]

In Appendix A, we present the same figures but using earnings (income from paid and self-employment) rather than market income. Earnings includes the salaries that firm owners pay to themselves but misses any capital related income. For the Incorporated cluster, mean earnings flattens out after the first few years (after the members of the cluster leave paid employment for the most part) and falls down to or below mean earnings for the Employed cluster. This highlights the importance of having data that captures capital income since without it, one might conclude that incorporation is associated with slower income growth. Interestingly, the shift to earnings from market income has virtually no effect on the height or path of income for the Self-Employed cluster. That is, self-employment has almost no associated capital income which, again, fits more with it being a version of an employment

[22]The patterns of income growth for the Incorporated relative to the Employed clusters for the NB are similar in nature to what Bhandari et al. (2025) show for workers and firm owners in the US.

[23]Similar results using longitudinal data are found in Georgarakos and Tatsiramos (2009) and Blume et al. (2009), all of which corroborates the distinction between incorporated and non-incorporated firms proposed in Levine and Rubinstein (2017).

state rather than the type of firm ownership reflected in incorporated firms.



(a) Immigrants (b) Native Born

Figure 7: Mean Market Income by Cluster, 2005 Cohort

Mean market income in each year, including zeros.

3.3 Income Before and After Entering Self-Employment

To this point, we have seen that the dynamic patterns that characterize our clusters imply that SE is similar in nature to an employment state, and one that is associated with lower income and inhabited by people with less education who enter through less selective entry classes. We can get a further indication of the nature of SE by tracking the income path as people move into SE and in the years after first opening an SE firm. Harju et al. (2025) use rich Finish administrative data to examine movements into incorporated firm ownership, showing that firm owners with high income open firms with more equity. They argue that this suggests that greater prior income captures a combination of greater ability and greater access to capital. They show that people opening incorporated firms are disproportionately likely to come from the top decile of the earned income distribution but that people from across the distribution get similar, substantial income gains from opening a firm.

We follow the approach in Harju et al. (2025) for both SE and incorporated firms, differentiating between our immigrant and NB cohort entrants but combining MICROINC and MAJINC firm ownership into one category (INC) for simplicity. More specifically, for the analysis of entry into SE ownership of the 2005 entry cohort, we compare individuals who were continuously in EMPL or NEMPL from 2005 through the year prior to a first movement into SE with those who remain in EMPL or NEMPL throughout the entire 2005–2019 period. We construct a similar comparison group for 2005 entrants who move into INC status but we note from our clustering exercise that very few paths in the Incorporated cluster involve any

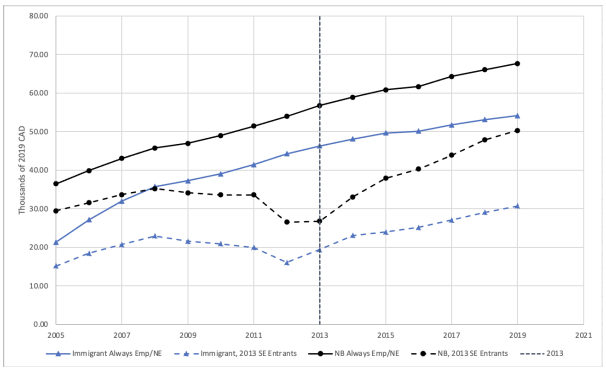
non-employment. Given that, we include only on people in the EMPL state throughout the period in the comparison group in order to focus on people who are more similar to incorporated firm owners. In both cases, the transition to opening a firm can occur at any point between 2008 and 2019, resulting in a staggered setting with up to twelve post-transition and fourteen pre-transition periods. We do not condition on individuals remaining in SE or INC after the transition year, allowing for the possibility that firm ownership is a stepping stone to other states. Note that in this exercise we are tracking all entrants into SE and INC regardless of which of our clusters they would be placed in.

Our implementation involves estimating the following event-study regression separately for immigrants and NB individuals, and for each of the two transition types (SE and INC):

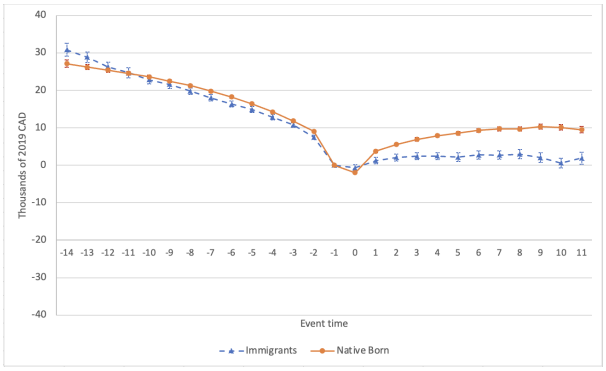
$$y_{it} = \alpha_i + \gamma_t + \sum_{\tau \neq -1} \beta_\tau \cdot D_i \cdot \mathbf{1}[t - t_i^* = \tau] + \delta X_{i,t} + \varepsilon_{it}, \tag{1}$$

where y_{it} is the real market income of individual i in year t, D_i is a dummy equal to one if individual i moves into SE (or INC) at any point between 2008–2019 and zero otherwise, t_i^* denotes the calendar year of the first transition into firm ownership of individual i, α_i and γ_t are individual and year fixed effects, and X_{it} includes a quadratic in age and province fixed-effects. Note that because we are following one cohort of labour market entrants, calendar year is the same as years since entry minus 2004. Standard errors are clustered at the individual level. β_τ is the coefficient of interest, capturing the average difference in the market income between individuals who transition into SE (or INC) and their corresponding comparison group of workers who do not open a firm, τ years relative to the year of transition. Because selection into SE or INC is not random, these estimates are descriptive in nature rather than causal. Indeed, the point of these "event studies" is to highlight the selectivity of the people who open firms in terms of their income paths. While our approach is similar in nature to the one used in the literature on identifying causal effects using panel data with staggered entry into the treated state, we do not face the identification problems studied in that literature for two reasons: 1) we are not trying to identify causal effects; 2) our control group consists only of never-treated individuals, eliminating potential problems with using previously treated individuals as controls.(see Harju et al. (2025) for more on this point)

In order to obtain a complete picture of what underlies the differenced patterns from our main specification, in the left panel of Figure 8 we plot the mean real market income levels by year for both the comparison group and for the set of people from the 2005 cohort who first open a firm in 2013. We just present one firm opening year for simplicity and choose 2013 as a typical middle year in the experience of the 2005 cohort. We present mean incomes



(a) 2013 SE Entrants and Emp/Nemp Comparison Group, Immigrants and Native Born



(b) Market Income of SE Entrants minus Comparison Group Relative to SE Entry Year

Figure 8: Market Income Paths Around SE Entry, 2005 Cohort

separately for the immigrant and NB 2005 cohorts. The paths show the clear superiority of market income for the NB, regardless of whether or not they open an SE firm, with the difference averaging around 23% after 5 years after entry for the comparison groups. They also show continuous gains with more experience for the non-firm owners while for those who eventually open an SE firm, their market incomes decline between year 4 after arrival and the year before owning an SE firm (at year 9). After starting firm ownership, market income growth roughly matches that for the non-firm owners among immigrants but is stronger for the NB firm owners.

The right panel of Figure 8, shows the differences in income growth between firm owners and non-owners with the differences normalized to zero in the year before initial firm ownership. Note that this utilizes the data in the left panel along with similar data for other firm ownership start years. The plotted difference series show that SE firm owners are on a declining market income path relative to non firm-owners as they approach the year of starting firm ownership. Interestingly, the declining path, measured in proportion to market income in the year just prior to firm ownership, is very similar for immigrants and non-immigrants. Both, in particular, show a sharp decline between year 2 and year 1 before starting firm ownership. After firm ownership, market income growth for firm owners and non-owners is very similar among immigrants while for NB, income growth is stronger for SE owners. These patterns fit with the narrative of SE being a state of last resort, with those who enter it being lower earners who are on a long term negative relative income trend that turns sharply worse just before starting firm ownership. At the same time, it appears rational for these people to enter SE firm ownership because entry is associated with a cessation in the downward trend in relative market income for immigrants and a reversal of that trend for the NB. The better performance for the NB fits with the cluster analysis showing that

SE was less likely to be an absorbing state and more likely to lead to INC ownership for non-immigrants than immigrants. That is, while SE has generally negative associations for immigrants, it is more mixed for non-immigrants.

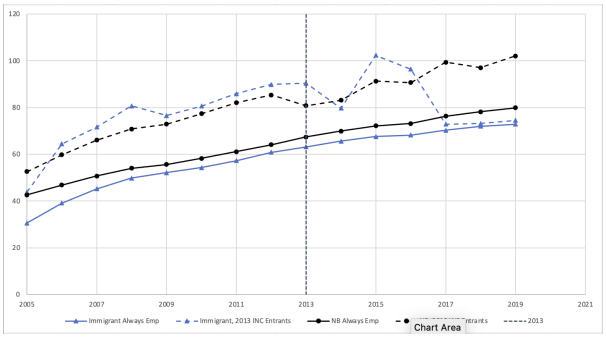
In Appendix B, we present the results from the estimation of a flexible competing risks model in which we examine time until first entry into either MAJINC, MICROINC, or SE. We assume new entrants from a cohort start in a combined state of either EMPL or NEMPL. We estimate the model pooling together our 2005 through 2016 entry cohorts and include a complete set of entry cohort, origin, and current location fixed effects. The only other covariates are the elements of a set of variables that take advantage of our rich tax data to capture earnings and employment paths. In particular, we calculate the earnings quintiles for all workers (immigrant and non-immigrant pooled) for each year and then assign workers to a 6 by 6 grid defined by those quintiles and a state of not working (i.e., being in either NEMPL or OUT). We include a complete set of 35 dummies corresponding to all but one of the cells in a grid defined by the combination of the states occupied in the previous two years, thereby capturing any earnings path in those two years. The results of that exercise show that being non-employed in a year after having low to medium earnings in the previous year positively predicts entry into the SE state, while having earnings in the top three quintiles in a year, regardless of prior earnings, is a strong negative predictor of entering SE. Again, poor labour market performance is associated with opening an SE firm.

3.4 Income Before and After Entering Incorporated Firm Ownership

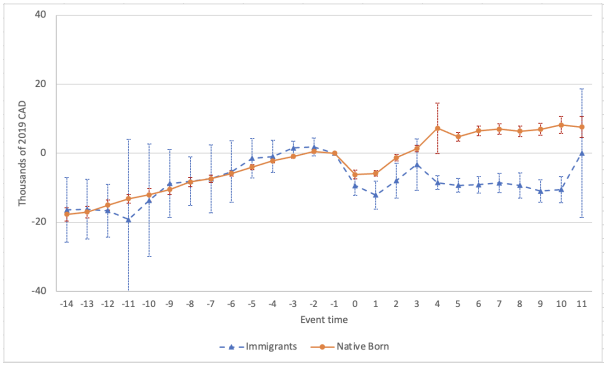
Figure 9 shows the market income paths related to entry into incorporated firm ownership (including both small and large incorporated firms). In contrast to SE owners, INC firm owners are positively selected, with higher market income both before and after starting firm ownership apart from the last few years for immigrant firm owners for this particular entry into firm ownership year group. Firm owners are also positively selected in terms of income growth. This is evident in the right panel containing the event studies plot. Again, we see that the pre-ownership income trends for owners versus non-owners are very similar for immigrants and non-immigrants, with both being positive in this case. This fits with the results in Harju et al. (2025) who also show positive relative growth in income for incorporated firm owners in the years before firm opening. Like them, we also find a slowdown in income growth just after firm opening, presumably while the owners plough earnings back into the new firm rather than taking them as income. It is noteworthy that the

slowdown in income for the SE owners happens just before firm opening, indicating problems in the paid labour market, while for INC firm owners they occur after firm opening.

After firm opening, the income patterns for immigrants and the NB part ways. After the first two years, NB market income resumes growing and, as in Harju et al. (2025), grows faster than for non-owners.[24] However, apart from the last year (which has a large associated confidence interval) the market incomes of immigrant INC firm owners do not recover from the initial, post-opening relative income loss. Measured in terms of their own incomes, firm performance is worse for immigrant than non-immigrant incorporated firm owners.[25]



(a) 2013 INC Entrants and Emp Comparison Group, Immigrants and Native Born



(b) Market Income of INC Entrants minus Comparison Group Relative to INC Entry Year

Figure 9: Market Income Paths Around INC Entry, 2005 Cohort

4 Incorporated Firm Performance

4.1 Firm Size

The weaker income results for immigrants compared to the native born after opening incorporated firms suggests that immigrant owned firms under-perform compared to those owned by the native born. To the extent this is true, this is important not only for the immigrants themselves but also for receiving economies which admit some immigrants in anticipation

[24]The fact that we are using market income, which includes dividends and other returns to capital is important for this result. If we only use employment earnings, including earnings paid by firm owners to themselves, non-immigrant owners show negative post-ownership growth.

[25]In Appendix A, we replicate the exercise in Harju et al. (2025), examining income paths around firm ownership separately by prior income quartile. Like them, we find similar income growth paths after firm opening regardless of prior income quartile, but while they show income growth after firm opening relative to a comparison group of non-owners, we find no extraordinary post-opening growth for firm owners. This is broadly true for both immigrants and the NB.

of them opening successful firms. The firms that governments are presumably hoping that immigrants will create are ones that exhibit the characteristics of a successful entrepreneurial venture - they grow in size in terms of both revenue and employment, have high productivity, and invest in capital. There is reason for hoping immigrants will disproportionately open such firms, including evidence that immigrants are less risk averse than the native born (Jaeger et al. (2010),Gibson and McKenzie (2011)),Dustmann et al. (2023))and so may be more willing to take the risk of opening dynamic ventures. Moreover, there is both anecdotal evidence and evidence from specific sectors that immigrants open successful firms (e.g., Hunt and Gauthier-Loiselle (2010), Brown et al. (2019), Peri et al. (2015), Blit et al. (2020) Saxenian (2002)). Our income path results, however, suggest that any such advantages may not be widespread. Our goal is to understand the differences in firm performance between immigrants in general and the native born.

We begin, in Figure 10, by depicting the evolution of the distribution of firm employment in incorporated firms owned by entrants in our immigrant and NB 2005 entry cohorts using survivor functions associated with those distributions.[26] To understand the survivor functions, consider the function for the immigrants in their first full year in Canada in the upper left panel. The first point at the left end of the plotted line shows the proportion of all immigrants who own an incorporated firm of any size (0.021). The second point, moving to the right, shows the proportion with at least one part time employee (0.009), and each point thereafter gives the proportion of all entrants who own an incorporated firm with at least that many employees. The difference between any two points reveals the proportion of entrants with exactly the number of employees listed at the left-most point. The advantage of examining survivor functions is that, at the same time, they capture both differences in the tendency to open any firm and the size distribution conditional on opening a firm. Thus, if immigrants open fewer firms with 4 or more employees conditional on opening any firm but open enough more firms, in total, such that as a proportion of all entrants they own more firms with at least 4 employees, then this would be captured in the function. And, indeed, some of the advantage of the immigrant entrants relative to the overall NB population in owning middle sized firms is of this nature.

Tracking the immigrant and NB cohort entrepreneurs across years reveals a pattern in which they are very similar in the size of the firms they own in the first year. In subsequent years, however, immigrants own more incorporated firms and the relative size distributions

[26]The firm tax data contains the total wage bill and the number of T4's issued but not hours of work. Statistics Canada creates a measure of the number of workers in Average Labour Units (ALUs) which are constructed by dividing the total wage bill by the average annual earnings per worker within cells defined by a firms' industry, province, and number of employees. Thus, the ALU approach counts the number of 'typical' workers a firm employs, where typical is defined by industry, province and raw number of workers.

changes substantially. In particular, the survivor function for the immigrants cuts that for native born entrants from above at '4+' employees in both 2014 and 2019. This implies that immigrants own more firms with fewer than 2 workers, both immigrants and the native born have the same probability of owning a firm with 4+ workers, but the native born have higher probabilities of hiring 10+ workers and 25+ workers. Thus, the immigrant advantage in opening incorporated firms is entirely about their opening more small firms. The result that immigrant firms are smaller on average fits with the US literature (Fairlie and Lofstrom (2013), Kerr and Kerr (2020)) and with results in Ranasinghe and Shao (2026) for Canada. The patterns here show that this is due to their owning more small firms and fewer large firms as opposed to, for example, their owning many more small firms but also more very large firms.

In a recent paper, Azoulay et al. (2022) claim that immigrants create more jobs (as entrepreneurs) than they 'take' (as workers). We can assess that claim by computing the proportion of jobs created by entrepreneurs in an entry cohort relative to the total number of immigrants in the cohort. We can calculate that proportion by multiplying the proportion who own a firm with employees (a MAJINC firm) by the mean number of workers per firm. For our 2005 entry cohort, that proportion is 0.26. That is, immigrant entrepreneurs created far fewer then enough jobs to employ the other workers who entered with them. This is not surprising. A large share of jobs in an economy are in large, well-established firms and we wouldn't expect new entrants to create such firms in 15 years. But it highlights the point that immigrant inflows do not generate an obvious increase in the employment rate through the entrepreneurs embedded in them - at least, not in the short to medium term. Moreover, it removes a possible explanation for the common finding (reaching back to Card (1990)) that immigrants have little effect on the mean wage in the receiving economy: that the immigrant inflow carries enough entrepreneurs and capital with it that it effectively just scales up the receiving economy without changing fundamental ratios that affect wages and vacancy rates. By way of comparison, the proportion of jobs to total entrants for our native born cohort is higher - 0.53 - because of their much larger firm size. Thus, also measured by total job creation per capita, the immigrant entrepreneurial contribution is lower.

The differences in the distribution of the number of employees also shows up in firm revenues. In Figure 11, we plot the percentiles of the distributions of firm revenues for 'active' incorporated firms (i.e., firms that do not have zero profits) for the immigrant and native born 2005 entry cohorts. The 10th percentile for both groups is very low and does not change substantially with time. The median revenue for native born firms is slightly higher than that for immigrant firms in all years, though neither grows noticeably. The real distinction is in the upper tail. The 90th and 95th percentiles of the distribution of firm

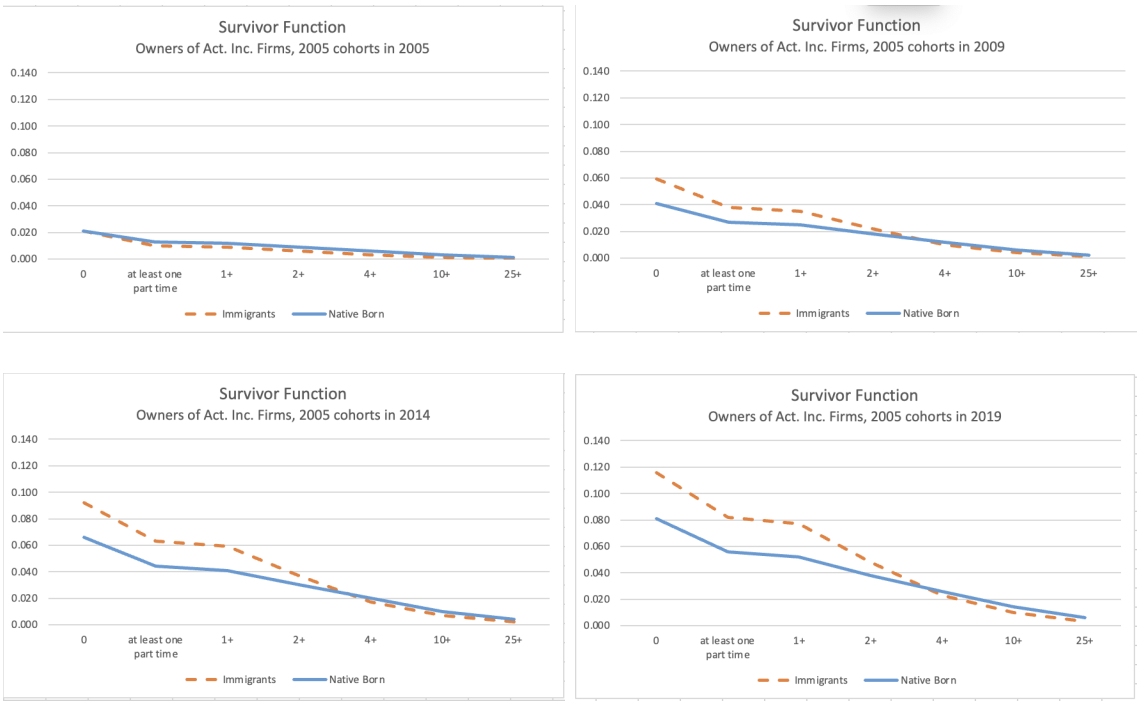


Figure 10: Survivor Functions for Firm Employment, Incorporated Firms, 2005 Entry Cohorts in Various Years

revenues are much larger for native born than immigrant firm owners just after entry and grow faster with time since entry. By 2019, the 90th and 95th percentiles of firm revenues for firms owned by the native born in the 2005 cohort are 2.7 and 3 times larger than the corresponding percentiles for firms owned by immigrants in the same cohort. The result that non-immigrant owned firms are larger in terms of both revenues and employment matches what Fairlie (2012) finds in cross-sectional results in the U.S.

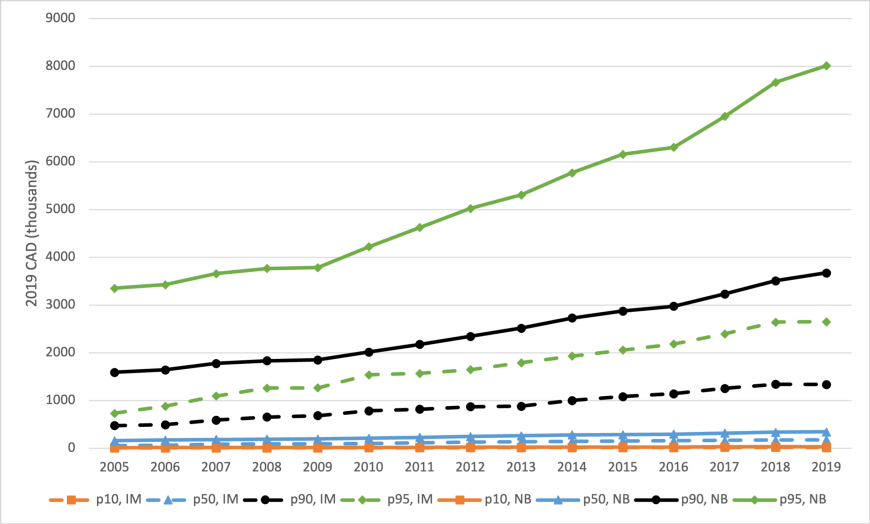


Figure 11: Percentiles of Total Revenue for Active Inc. Firms: 2005 Immigrant and Native Born Cohorts

A plausible explanation for the differences in firm size (and, potentially, other differences in firm performance) comes from differences in industrial composition between immigrant and native born firms. Comparing the 2005 immigrant and NB cohorts, by 2019, immigrant incorporated firms were more likely to be in Wholesale or Retail Trade and Accommodation and Food Services (19.6% versus 14.7%) and in Professional/Technical Services (21% versus 15.6%) with offsetting lower probabilities of being in Construction (9.7% versus 16.5%).[27] To investigate the role of industrial composition, we construct a counterfactual for mean employees per firm in which we use the mean number of employees per firm in each two digit industry from Immigrant owned firms but weight them by the proportion of NB firms in each industry in each year. The result is presented in Figure 12. We, again, see that NB incorporated firms are larger and show more growth. But the key point is that the counterfactual series is very similar to the Immigrant series, implying that differences in industrial composition do not explain the size differences between Immigrant versus NB firms. This arises largely because the NB firm advantage exists within virtually all sectors. In Appendix A, we present results for the same counterfactual exercise for Revenues and Value Added per Worker. Those, too, show that industrial composition differences are not the source of differences in firm performance.

[27]The complete industrial distributions for immigrant and NB incorporated firms for the 2005 entry cohort over time are presented in Appendix A.

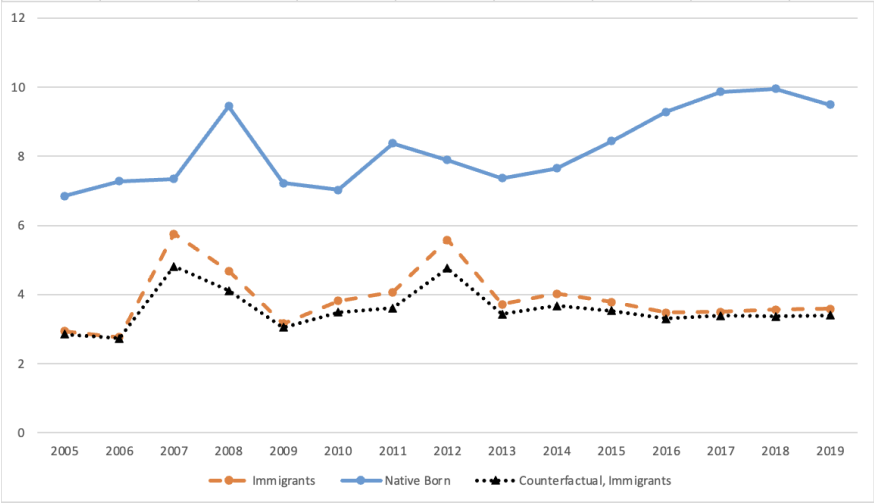


Figure 12: Mean Number of Employees for Active Inc. Firms: 2005 Immigrant and Native Born Cohorts and Counterfactual

4.2 Firm birth, growth and death

Underlying these movements in firm size are dynamics described by birth, growth, and death. Understanding how these differ between immigrants and the native born can provide some insight into whether immigrant firms are special. We saw some of this in our cluster analysis - movements from owning MICROINC to MAJINC firms is similar for both the immigrant and native born cohorts. Here, we will investigate potential differences in greater detail.

We divide MAJINC firms into: Small (2 to 4 employees); Medium (5 to 9 employees); and Big (10 or more employees) firms. We compute transition rates among these firm size states and between these states and the combination of all other states (EMP, NEMP, OUT, SE, and MICROINC). The goal is to understand the growth of what we view as potentially entrepreneurial firms - incorporated firms with 2 or more employees. In the top left panel of Figure 13, we plot new firm creation in a year as a proportion of people in one of the non-MAJINC states in the year before. We average these proportions across five year ranges for confidentiality reasons. The figure shows that immigrants create Small firms at a higher rate than the native born but the creation of Medium and Big firms are generally the same. The top right panel shows the probability of a firm of a given size being observed as the next larger size in the following year. Here, the native born show higher rates for movement from Small to Medium and similar rates of moving from Medium to Big to immigrants. The bottom left panel shows the probability a firm in a given size category is observed in the next smallest category in the following year. Immigrant owned firms have higher rates of shrinkage starting both from Medium and Big sizes. Finally, in the bottom right panel we

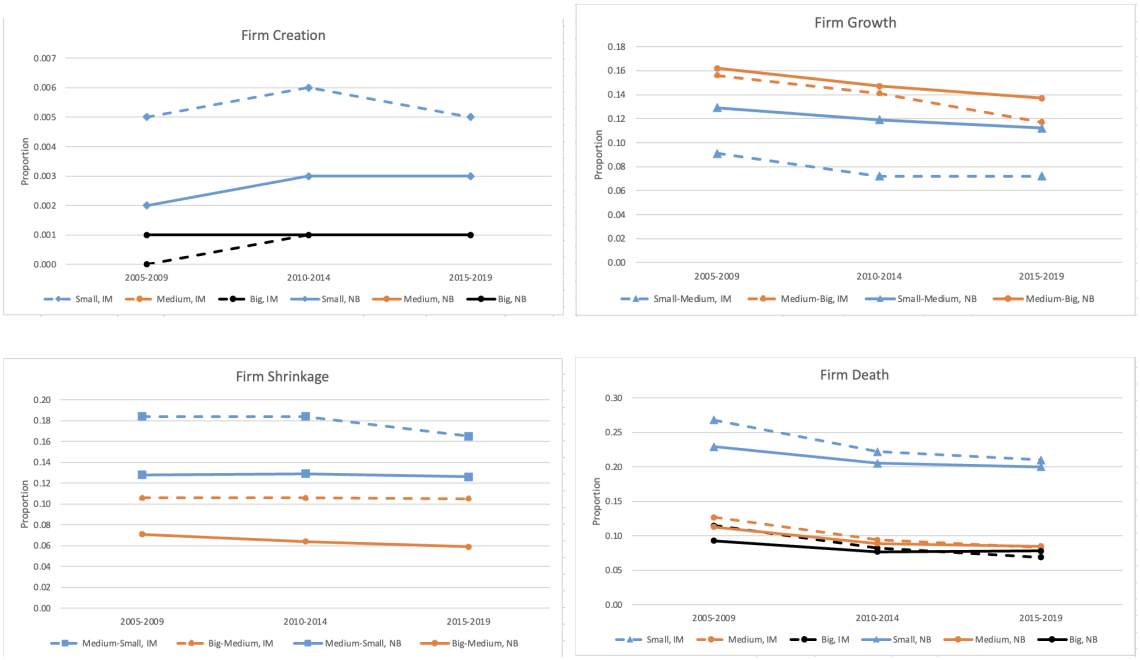


Figure 13: Firm Birth, Growth, Shrinkage, and Death Rates: 2005 Immigrant and Native Born Cohorts

plot the proportion of firms of each size that cease to be a MAJINC firm by the following year. Immigrant firms show somewhat higher death rates in all categories.

Taking this information together, the fact that immigrants own more Small firms, as we saw in the survivor functions, arises because they create more Small firms and those firms are less likely to grow combined with the fact that the Medium firms they do create are more likely to shrink. In general, immigrant firm dynamics show more churning (more birth, death and shrinkage). More churning is sometimes viewed as a sign of entrepreneurial activity as entrepreneurs try out ideas some of which fail, but with immigrant firms having lower growth rates, that interpretation doesn't appear to fit in this case. Instead, the dynamics processes point to immigrant firms being more unstable in a way that keeps them smaller than their native born owned counterparts. Some of this may have to do with differences in the industrial composition of firms, with immigrants more likely to open firms in retail and accommodation and food services, though they are also more likely to open Professional and Technical sector firms (which may be more stable), and the native born are more likely to open construction firms (which may be less so). We present evidence on the industrial composition of firm ownership in Appendix A.

4.3 Productivity and Investment

Regardless of size, of course, creating firms that are particularly productive or generate considerable investment could also be taken as signs of immigrant success. We measure productivity as value added per worker (VAPW), where value added is Statistics Canada's definition: capital income plus the wage bill.[28] We present percentiles of VAPW for our immigrant and native born cohorts for incorporated firms with 1 or more workers in Figure 14. By this measure, both productivity and the variance of the productivity distribution grows with time - i.e., there is a set of more innovative firms that emerges with time. As with revenues and employment, the native born distribution is a rightward shift of the immigrant distribution in almost every year. For VAPW, though, the right tails of the two distributions are quite similar for the first 8 years, after which the native born firms pull away. That is, if the upper tail of the VAPW distribution represents the most innovative firms, it is the new native born firms that grow more in that innovation in the long run. It is worth noting, as a comparison, that value added per worker in Canada in 2019 was $110,000.[29] The median VAPW values for our immigrant and native born cohorts are below that number (as is the 75th percentile of the immigrant distribution) while the 90th percentile is above it. If we could manage to admit only the most entrepreneurial individuals, then increased business immigration would improve overall productivity, even if less than new native born entrants. With current selection, though, this isn't the case.

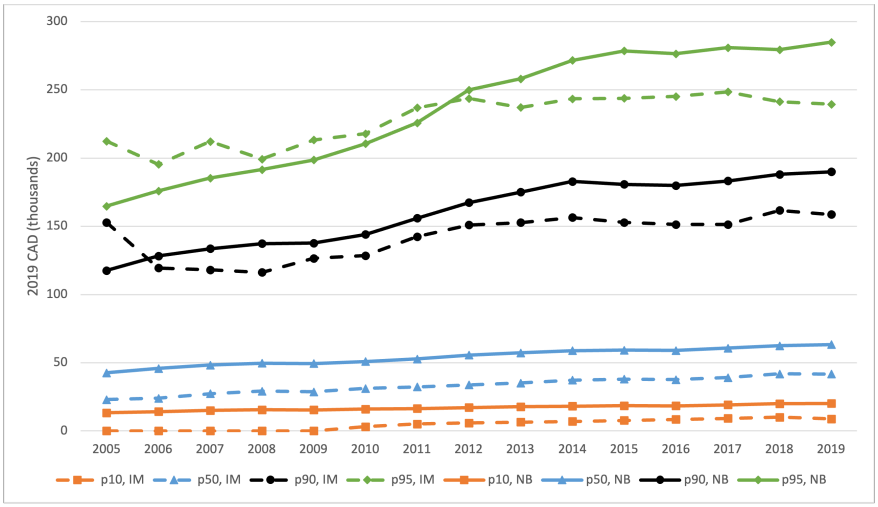


Figure 14: Percentiles of Value Added Per Worker for Owners of Active Inc. Firms with 1 or More Workers: 2005 Immigrant and Native Born Cohorts

[28]For all per worker figures, we restrict the sample to firms with at least one worker.

[29]Statistics Canada Table 36-10-0480-01, `https://www150.statcan.gc.ca/t1/tbl1/en/cv.action?pid=3610048001`. Accessed March, 2025.

In Figure 15, we present evidence on firm investment and capital stock. In the left top panel, we plot the proportion of firms in the immigrant and native born cohorts who have positive investment in each year.[30] In the right top panel, we plot average investment conditional on making positive investments (the lines) along with the component types of investment (the bars). Mirroring the other measures of firm performance, native born firms are both more likely to invest and, conditional on investing, invest more - with both gaps growing with time since entry. It is also noteworthy that investment in machinery and equipment makes up more than 50% of investment in nearly every year for native born owned firms but is always under (and, typically, well under) 50% for immigrant owned firms, where investment in buildings makes up the lion's share. The biggest differences in these shares occur toward the end of the series when the native born - immigrant firm investment gap is growing. That is, the difference when the native born firms pull ahead in amount invested is due to their larger investment in machinery and equipment. Investment in buildings seems less like innovative investment than in equipment. The bottom panel shows that the differences in investment translate into differences in capital per worker (with capital measured as total tangible assets). Native born firms have substantially higher capital - labour ratios at the 90th and 95th percentiles of the distributions.

It is interesting to consider the investment and VAPW outcomes together. It seems possible that the lower VAPW in immigrant owned firms could be accounted for by lower capital stock arising out of inferior investment. In that case, lower VAPW in immigrant owned firms might not reflect differences in productivity but differences in access to capital markets.

To examine this, we compute the change in Total Factor Productivity (TFP) at firm i as:

$$\Delta TFP_{it} = \Delta \ln VA_{it} - \alpha_1 \Delta \ln K_{it} - \alpha_2 \Delta \ln L_{it} \tag{2}$$

where VA_{it} is value added, K_{it} is the capital stock (based on data on Tangible Assets at the firm), and L_{it} is total employment. This assumes a Cobb-Douglas production function. We use standard capital and labour shares: α_1=0.33 and α_1=0.67. In Figure 16, we plot percentiles of this series for our immigrant and native born cohorts. Those percentile indicate that immigrant owned firms had better TFP growth in the upper tail of the distribution in the early years after entry but from approximately year 6 onward the two distributions are substantially similar. It is worth noting that our TFP measure is conditional on the firm having a valid capital measure and, as a result, will tend not to include firms that don't

[30]Investment is the sum of spending on buildings, machinery and equipment, intangibles, and 'assets without classification'.

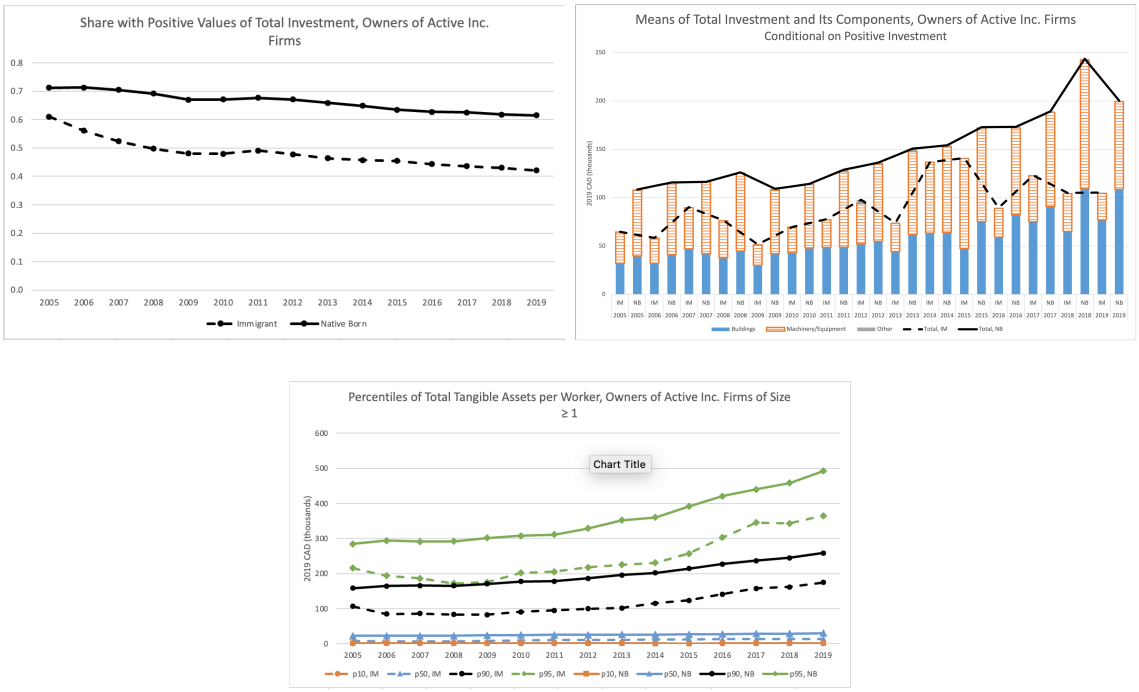


Figure 15: Measures of Firm Investment and Capital per Worker, Incorporated Firms, 2005 Entry Cohorts in Various Years

invest - of which there are more among immigrant owners. Thus, the true productivity differences between immigrant and native born firms is probably better represented by the VAPW measure where we can include all firms. Still, the fact that immigrant and native born firms are equally productive in a TFP sense, i.e., holding capital differences constant, suggests that immigrant incorporated firm owners are as entrepreneurial as the native born owners but are hampered by a lack of investment.

These results are related to a literature on the financing of ventures by immigrant entrepreneurs. Some articles in that literature argue that lesser access to formal borrowing channels is an impediment to immigrant firm ownership and firm growth (e.g., Kushnirovich and Heilbrunn (2008)). But more recent evidence from the U.S. indicates that immigrant and non-immigrant owned businesses are actually quite similar in their sources of start-up financing, with a considerable amount coming from existing assets (Fairlie (2012),Fairlie and Lofstrom (2015)). Ostrovsky et al. (2019), using Canadian survey data, conclude that immigrants and non-immigrants finance their businesses quite similarly, though they do show some evidence that immigrants are less likely to obtain financing from formal financial institutions and somewhat less likely to have their credit applications approved. Our approach differs from earlier papers in our use of a control group of the native born who are also

just becoming active in the economy. Relative to that comparison group, new immigrants appear to face greater investment constraints and that, in turn, results in their poorer firm performance.

The literature on relative economic performance across economies argues that the average revenue product of capital (ARPK) and the average revenue product of labour (ARPL) can be used to determine whether firms in one economy face greater frictions in the capital and labour markets than in other economies. The argument is that frictions amount to an extra cost per unit of the input, causing firms to reduce their use of the input and move farther back up the marginal revenue product schedule.(Restuccia and Rogerson (2008), Hsieh and Klenow (2009)). Ranasinghe and Shao (2026) adapt this insight to their examination of entrepreneurs in Canada, investigating whether immigrant firm owners have higher ARPK and ARPL values and, therefore, face greater restrictions in the capital and labour markets. Controlling for industry, time, province, gender, and age effects, they estimate significant positive differences for immigrant owned firms for both ARPK and ARPL. We implemented their specification with our data and, similarly, obtain a positive and statistically significantly higher ARPK value for immigrant compared to native born owners.[31] Ranasinghe and Shao (2026) show that the immigrant-NB difference in ARPK declines with the age of the firm, which fits with immigrants getting better access to capital or more constrained immigrant firms dying early. In Appendix A, we present figures showing ARPK for immigrant and NB firms by industry. We also show the standard deviation of ARPK for immigrant and NB firms by industry. The standard deviation is larger for immigrants, implying that they have more variable access to capital markets. Altogether, this evidence fits with the story that immigrants have more restricted access to capital markets. Ranasinghe and Shao (2026) calibrate a general equilibrium model that incorporates immigrant firms facing greater frictions in the capital and labour markets and use it to assess the impact of policies that would reduce those frictions.[32]

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[31]ARPK is defined as total revenues divided by total tangible assets. Their estimated coefficient on an immigrant ownership dummy in a regression with the log of ARPK as the dependent variable is 0.145 (se 0.002) while our estimate is 0.268 (se 0.016). The difference may be due to the fact that they restrict their sample to single establishment firms while we do not).

[32]They estimate a coefficient of 0.107 (se 0.001) on the immigrant ownership dummy variable in a regression of the log of ARPL on that dummy and industry, year, province, gender, and age effects. We obtain an estimate of -.275 (se 0.009) in the same regression. Thus, if anything, we find that immigrant owned firms over-use labour. The difference stems from their using the wage bill as the denominator for ARPL while we use employment. Using the wage bill in principle weights workers by their efficiency, as reflected in their wage, but we are worried that differential wages for immigrant workers may partly reflect discrimination rather than efficiency differences and immigrant owned firms hire more immigrant workers. For that reason, we believe employment is preferable as the measure of the labour input.

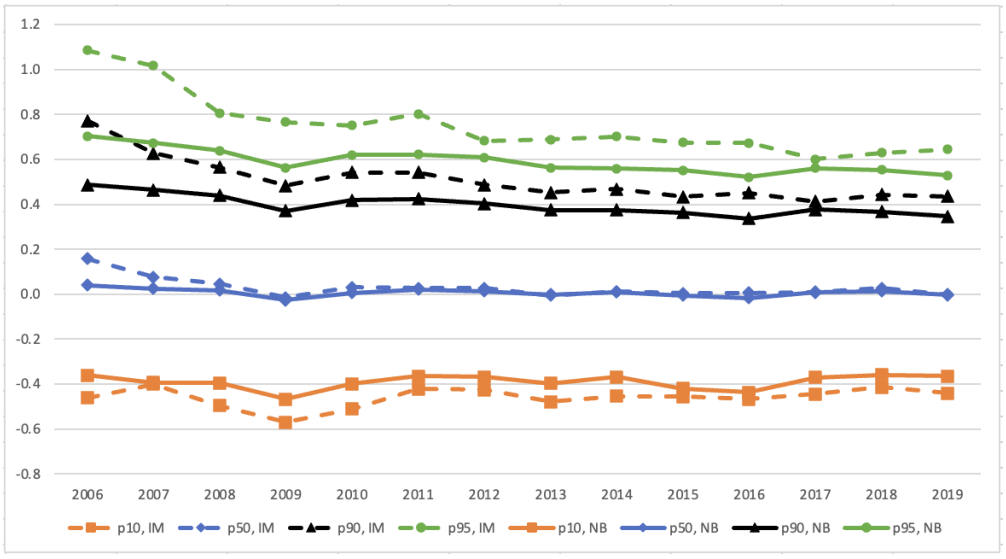


Figure 16: Percentiles of TFP Growth, Owners of Active Inc. Firms with Valid log VA, log K, log L: 2005 Immigrant and Native Born Cohorts

5 Conclusion

Many immigrant receiving countries have components of their immigration systems that target entrepreneurs. This is done, presumably, in the hope that immigrant entrepreneurs will open firms and create jobs in the host economy - a hope that gets some support in the observation that a disproportionate number of firms in Silicon Valley were started by immigrants (Saxenian (2002)). In this paper, we use rich Canadian administrative data to examine the process by which immigrants enter into firm ownership. Matching earlier work (e.g., Fairlie and Lofstrom (2013) and Schuetze and Antecol (2005)), we find that soon after arrival, immigrants are more likely than the native born to own firms. But we also find that this advantage is largely accounted for by the ownership of nonincorporated firms. Those firms rarely have employees and, as others have observed, tend to look more like locations of last resort than innovative firms (Levine and Rubinstein (2017),Blume et al. (2009)). In our data, immigrants who start nonincorporated firms tend to be lower educated, to have recently experienced a spell of unemployment or low earnings, and to have low income relative to paid employees or owners of incorporated firms when running their firm. Moreover, nonincorporated firm ownership, for many, is an absorbing state. It is very rarely a stepping stone to opening an incorporated firm with employees - the kind of firm that governments are hoping that immigrant entrepreneurs will open.

At the other end of the spectrum are incorporated firms with employees, which Levine and

Rubinstein (2017) argue are more likely to be the locations of innovative activity. Immigrants have higher rates of owning incorporated firms with no or one employee than the native born but quite similar rates of owning an incorporated firm with at least two employees. However, when we investigate the nature of those incorporated firms we find that immigrant owned incorporated firms are smaller, are more likely to die or shrink, have lower revenues, investment and value added per worker than incorporated firms opened by a matching set of young native born people. We do find that the distributions of firm TFP growth are very similar for firms owned by immigrants and the matching native born group. This suggests that immigrant entrepreneurs are as productive - though no more so - than native born entrepreneurs once we take account of differences in capital in the firm. A possible policy implication is that there could be benefits to helping immigrants get better access to capital to invest in starting or expanding firms.

So, the answer to the question posed in our title is no - immigrants do not appear to be much more entrepreneurial than the native born in the sense of opening successful firms that look 'entrepreneurial' (incorporated firms with employees), and on many measures they appear to be less entrepreneurial. The combination of this result with the predominance of owning nonincorporated firms by immigrants implies that firm ownership is not a major element of success in immigrants adapting to their new economy. The option of starting a self-employed firm may prevent worse outcomes in the paid workforce but it is not a part of a strong income growth path. In terms of contributions to the receiving economy, the number of jobs associated with immigrant owned firms is far fewer than the number of immigrant workers in the cohorts we examined.

The one caveat to these conclusions might be found in the literature on immigration and innovation in the tech and science sectors. A set of papers argues that highly educated immigrants in STEM fields bring contributions in terms of innovations and the opening of tech firms (Hunt and Gauthier-Loiselle (2010), Brown et al. (2019) Picot and Ostrovsky (2017), Peri et al. (2015), Saxenian (2002), etc.). However, Blit et al. (2020) find minimal impacts on patenting in the Canadian context except when the STEM educated immigrants are employed in STEM fields. Our results are more closely aligned with theirs but, more generally, are about firm ownership rather than the type of innovation captured in patenting, which does not have to involve firm creation.

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